Joint Polygamma Moments, Gap-Eight Reconstruction, and Multiscale Stieltjes Resonances

Research continuation for the ProveIt project

12 October 2026

Abstract

We advance three explicitly open directions in the current ProveIt polylogarithm and special-function reports. First, an unpaired deformation of a symplectic Gamma integral separates every joint polygamma cumulant whose total derivative weight is at most $2n+2$. Positive-order statistics obey an exact finite Wick rule, whereas the digamma sector has explicitly nonzero higher cumulants. A small-parameter limit proves that the Gaussian range cannot be extended universally. Second, we make fixed-gap harmonic tail reconstruction effective for every prescribed even gap and complete gap eight for all indices. The proof combines explicit Gamma-integral asymptotics with an exact finite-field certificate for the remaining 8187 matrices; it also produces an all-order family of ordinary, absolutely convergent polygamma-product sums equal to $-1/2$. Third, we evaluate the constant term in collisions of arbitrary argument derivatives of generalized Stieltjes constants at two positive monomial scales. Three elementary resonance equations describe all new contacts. Their coefficients reduce to zeta derivatives, harmonic numbers, and Stieltjes constants at one. A first-index specialization realizes every integer trigamma tail as a collision contact, with a polylogarithmic generator and explicit primitive. Barnes-integral companion evaluations, source qualifications, reproducible verification programs, and a precise research agenda accompany the proofs. Novelty is asserted relative to the inspected repository baseline, with classical ingredients identified.

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1 Scope, principal results, and conventions

The starting point is the ProveIt repository at revision `03112a7d3b91` (full revision in [1]), inspected on 12 October 2026. We read the polylogarithm manuscript and the text sources in the incoming reports available at that revision; the most immediate predecessors are the ordered-boundary, gap-six, and Gamma-normalization reports [1, 2, 3, 4]. This matters because several questions suggested by older manuscript passages have already been answered in newer incoming material. In particular, complete gap six, fixed-rate Stieltjes collisions, and the paired Gamma deformation are prior results, not new contributions of this article.

Our aim is to obtain exact evaluations and structural formulas. Estimates are used where needed to prove identities or to turn eventual nonsingularity into a theorem valid for all indices. They are not the principal conclusions. Each of the three main directions gives a finite, explicit way to evaluate an infinite family.

1.1 Three advances and their exact scope

Joint polygamma moments. For the positive n-variable Gamma density defined in (2.1), introduce $R_q = \sum_{j=1}^{n} \Re\psi^{(q)}(\sigma + ix_j/2)$. Theorem 2.1 evaluates all joint cumulants satisfying $\sum_i (q_i + 1) \le 2n + 2$. It gives means, covariances, and the particularly simple higher-cumulant formula

$$\kappa(R_{q_1}, \dots, R_{q_r}) = -2^{-r}\psi^{(r-1)}((2n+2)\sigma) \prod_i \delta_{q_i 0}, \qquad r \ge 3.$$

Thus the positive-order sector has exact Gaussian moments in a finite weighted range, while the digamma sector retains a non-Gaussian correction. This answers the joint-separation question of the latest Gamma report in that range. The partition formula (2.12) evaluates the corresponding ordinary integrals without leaving parameter derivatives unevaluated. Proposition 2.4 also proves a specific obstruction to extending the source's single-trigamma matching degree universally.

Gap-eight reconstruction and sums. For $p = 2b - 2m$, $q = 2b$, the harmonic-tail problem reduces to a rational matrix $M_{b,m}$. Theorem 3.3 gives a fully computable threshold $B(m)$ beyond which it is nonsingular, for every prescribed $m \ge 1$. Theorem 4.4 completes $m = 4$ at every $b \ge 5$, settling the explicitly open gap-eight ray. Its finite step is an exact computation, not a numerical extrapolation. Corollary 4.5 then gives rational polynomials P_b, S_b such that

$$\sum_{n=0}^{\infty} \Big[P_b(n + \tfrac{1}{2})\zeta(2b-8, n+1)\zeta(2b, n+1) - S_b(n + \tfrac{1}{2}) \Big] = -\frac{1}{2}.$$

The subtracted series is absolutely convergent. Matrix rank is taken on the indicated formal coordinates; it is not a claim that the evaluated zeta values are rationally or algebraically independent.

Different-power Stieltjes collisions. Theorem 5.3 evaluates the change in every finite-part constant when the two arguments are shifted by $a = \lambda\epsilon^{\alpha}$, $b = \mu\epsilon^{\beta}$, with positive parameters and unequal exponents. The equal-exponent case is inherited. The answer consists of two explicit mixed contacts and at most one singular–singular contact. The latter can survive precisely when

$$j(\alpha - \beta) = \beta(r + t + 1), \qquad j \in \mathbb{Z}_{\ge 1}, \quad \alpha > \beta.$$

The mixed contacts obey two further integer selection rules. This classification explains constants absent from the fixed-rate theorem, including the two renormalized limits $-1-\zeta(3)$ and $-1+\zeta(4)$

in (5.8)–(5.9). An especially short consequence is

$$\mathrm{CT}_\epsilon \int_0^1 \gamma_1(x+\epsilon^{j+1})\gamma_0(x+\epsilon^j)\,dx = \frac{3}{2}\gamma_2 + 2\gamma\zeta(2) - \zeta(3) + \psi'(j+1).$$

Here the fixed finite part is inherited; the additional trigamma tail is the newly identified collision term.

1.2 Classical ingredients and the novelty boundary

The multidimensional Gamma integral is a classical Gustafson integral; we use the explicit symplectic normalization reproduced in [6, Eq. (2)], together with the repository's determinant transformation. The new step is separation in independent power-sum coordinates through their exact injectivity range. The reconstruction proof builds on the repository's Bernoulli recurrence and Gamma basis. Its new steps are a uniform effective threshold, the complete gap-eight certificate, and the associated evaluated sums.

Finite-part integration and generalized Stieltjes-transform asymptotics have an established literature [8]; integrals of products of Hurwitz zeta functions also occur in [9]. The monomial collision theorem below uses elementary local subtraction, an incomplete beta expansion, and cancellation of two meromorphic branches. Its new content is the evaluated all-index contact law for the specified two-scale problem. The Barnes companion in Section 6 is explicitly presented as a consequence of the classical first Barnes lemma [7, 5.13.3]. We do not claim that parameter differentiation, Bell polynomials, Barnes integration, or the resulting elementary Euler-sum generators are new methods.

The comparison establishes advances over the inspected repository, not an exhaustive historical-priority determination. The claims are proved in the usual mathematical sense, with an explicitly isolated exact computational component for the finite gap-eight interval. The accompanying numerical calculations are independent diagnostics; they are not proofs by integer-relation recognition. No proof-assistant formalization is claimed.

1.3 Notation and finite-part convention

We use $\psi = \Gamma'/\Gamma$, $\psi^{(0)} = \psi$, and the standard Hurwitz zeta function $\zeta(s,x)$. A superscript in parentheses on $\gamma_m(x)$ denotes an argument derivative, whereas m is its Stieltjes index. The convention for generalized Stieltjes constants is

$$\zeta(1+u,x) = \frac{1}{u} + \sum_{m\ge 0} \frac{(-1)^m}{m!}\gamma_m(x)u^m, \qquad \gamma_m = \gamma_m(1), \quad \gamma_0 = \gamma.$$

Thus $\gamma_0(x) = -\psi(x)$. Rising factorials are $(a)_k$; falling factorials are explicitly underlined. We set

$$H_j^{(h)} = \sum_{k=1}^{j} k^{-h}, \qquad H_j = H_j^{(1)}, \qquad H_0^{(h)} = 0.$$

Bernoulli numbers have generating function $t/(e^t-1) = \sum_{k\ge 0} B_k t^k/k!$. All logarithms of positive arguments are real; analytic logarithms elsewhere are fixed locally at the stated base point.

The operation CT_ϵ extracts the coefficient of $\epsilon^0(\log\epsilon)^0$ from the specified power–logarithm expansion. It is a coordinate-dependent finite part, not an unspecified regularization and not necessarily a limit. We supply explicit ordinary subtractions when stating a renormalized limit. The monomial exponents are fixed while $\epsilon \downarrow 0$; a jointly varying exponent ratio near a resonance is a separate problem.

2 Joint Gamma–polygamma cumulants

The latest incoming Gamma-normalization report [4] evaluates the appropriate Gustafson determinant, establishes a paired parameter deformation, and proves Gaussian moments for a single trigamma statistic through degree $n+1$. Its research question Q2 asks for a precise classification of joint polygamma moments. The theorem below answers this question in a complete weighted range. It also includes the digamma statistic and all even polygamma orders, which require the unpaired deformation. The classical integral and the source's density are inherited; the explicit cumulant classification and its consequences are the present extension. No claim of priority over all literature is made.

For $n \geq 1$, $\sigma > 0$, put $N = 2n+2$ and consider the probability measure on $(0,\infty)^n$ with density proportional to

$$\Delta(x_1^2,\dots,x_n^2)\Delta(\cosh \pi x_1,\dots,\cosh \pi x_n)\prod_{j=1}^{n} x_j \sinh(\pi x_j)|\Gamma(\sigma + ix_j/2)|^{2N}, \tag{2.1}$$

where $\Delta(y) = \prod_{j<k}(y_k - y_j)$. Its normalizer is

$$Z_{n,\sigma} = 2^{n(3n+1)/2}\pi^{n(n+1)} n! \frac{\Gamma(2\sigma)^{(n+1)(2n+1)}}{\Gamma(N\sigma)}. \tag{2.2}$$

The Vandermondes have the same sign, so this is a positive measure. Write $\mathbb{E} = \mathbb{E}_{n,\sigma}$ and define, for $q \geq 0$,

$$R_q = \sum_{j=1}^{n} \Re\psi^{(q)}(\sigma + ix_j/2), \qquad \psi^{(0)} = \psi. \tag{2.3}$$

The *weight* of R_q is $q+1$. A joint cumulant means

$$\kappa(Y_1,\dots,Y_r) = \partial_{t_1}\cdots\partial_{t_r} \log \mathbb{E}e^{t_1Y_1+\cdots+t_rY_r}\Big|_{\mathbf{t}=0}.$$

All fixed moments exist. Indeed the positive-order polygammas are bounded on each vertical line $\Re z = \sigma$, the digamma grows at most logarithmically, and the Gamma weight has an exponential margin after multiplication by every fixed polynomial or logarithmic power.

Theorem 2.1 (Explicit joint cumulants in the complete weighted range)**.** *Let $q_1,\dots,q_r \geq 0$ and suppose*

$$\sum_{i=1}^{r}(q_i+1) \leq N = 2n+2. \tag{2.4}$$

The one-point and two-point cumulants are

$$\mathbb{E}R_q = \frac{1}{2}\left[(N-2^q)\psi^{(q)}(2\sigma) - \delta_{q0}\psi(N\sigma)\right], \tag{2.5}$$

$$\operatorname{Cov}(R_p,R_q) = \frac{1}{4}\left[\psi^{(p+q+1)}(2\sigma) - \delta_{p0}\delta_{q0}\psi'(N\sigma)\right]. \tag{2.6}$$

For every $r \geq 3$ in the same weighted range,

$$\boxed{\kappa(R_{q_1},\dots,R_{q_r}) = -\frac{\psi^{(r-1)}(N\sigma)}{2^r}\prod_{i=1}^{r}\delta_{q_i0}.} \tag{2.7}$$

Thus every mixed cumulant of order at least three that contains a positive-order polygamma statistic vanishes in this range.

2.1 Proof by independent parameter shifts

For independent parameters $z_1, \dots, z_N$ near zero, deform all Gamma parameters to $\sigma + z_\ell$. Gustafson's symplectic Barnes integral, in the normalization of Derkachov–Manashov [6, Eq. (2)], and the source's determinant-to-density transformation give

$$\begin{aligned}\mathbb{E}\prod_{j=1}^{n}\prod_{\ell=1}^{N}\frac{\Gamma(\sigma+z_\ell+ix_j/2)\Gamma(\sigma+z_\ell-ix_j/2)}{\Gamma(\sigma+ix_j/2)\Gamma(\sigma-ix_j/2)}\\ =\frac{\Gamma(N\sigma)}{\Gamma(N\sigma+\sum_\ell z_\ell)}\prod_{\ell<h}\frac{\Gamma(2\sigma+z_\ell+z_h)}{\Gamma(2\sigma)}.\end{aligned}\tag{2.8}$$

This is an identity of holomorphic germs. On a sufficiently small polydisc every shifted Gamma parameter remains in the right half-plane. Uniform Stirling bounds leave an integrable exponential majorant, even after any fixed mixed derivative; they justify coefficient extraction under the integral. Holomorphic log-Gamma is used on that half-plane, and the analytic logarithm of the expectation is fixed by its value zero at the origin.

Put $p_k = \sum_{\ell=1}^{N} z_\ell^k$. Taylor expansion of the logarithm of the integrand in (2.8) gives

$$\log\left(\prod_{j,\ell}\frac{\Gamma(\sigma+z_\ell+ix_j/2)\Gamma(\sigma+z_\ell-ix_j/2)}{\Gamma(\sigma+ix_j/2)\Gamma(\sigma-ix_j/2)}\right)=2\sum_{k\ge1}\frac{R_{k-1}}{k!}p_k.$$

For every integer $k \ge 1$ the elementary pair-sum identity is

$$\sum_{\ell<h}(z_\ell+z_h)^k=(N-2^{k-1})p_k+\frac{1}{2}\sum_{a=1}^{k-1}\binom{k}{a}p_ap_{k-a}.\tag{2.9}$$

The coefficient of p_k follows by excluding the diagonal from $\sum_{\ell,h} z_\ell^a z_h^{k-a}$; the two endpoint terms each contribute $(N-1)p_k$ before the factor $1/2$ is applied. Taking logarithms in (2.8) therefore gives

$$\begin{aligned}&\log\mathbb{E}\exp\left(2\sum_{k\ge1}\frac{R_{k-1}}{k!}p_k\right)\\ &=\sum_{k\ge1}\frac{(N-2^{k-1})\psi^{(k-1)}(2\sigma)}{k!}p_k+\frac{1}{2}\sum_{a,b\ge1}\frac{\psi^{(a+b-1)}(2\sigma)}{a!b!}p_ap_b-\sum_{r\ge1}\frac{\psi^{(r-1)}(N\sigma)}{r!}p_1^r.\end{aligned}\tag{2.10}$$

The next finite algebra observation is the step that permits separation of the different statistics, rather than merely recording a generating identity.

Lemma 2.2 (Independent power-sum coordinates through degree N)**.** *Assign weight k to an indeterminate P_k. Substitution $P_k \mapsto p_k(z_1, \dots, z_N)$ is injective on the space of polynomials of weighted degree at most N.*

Proof. Only $P_1, \dots, P_N$ can occur in that degree range. Newton's identities give an invertible triangular change between $p_1, \dots, p_N$ and the first N elementary symmetric polynomials, which are algebraically independent. Alternatively the Jacobian of the power sums is

$$\det\left(\frac{\partial p_k}{\partial z_\ell}\right)_{k,\ell=1}^{N}=N!\prod_{\ell<h}(z_h-z_\ell),$$

which is not the zero polynomial. Either proof gives the claimed injectivity. □

Proof of Theorem 2.1. The homogeneous component of degree $d \le N$ in (2.10) involves only $p_1, \dots, p_d$. By Lemma 2.2, compare its coefficients as polynomials in independent power-sum variables. Set $p_k = k! t_{k-1}/2$ in this finite-degree comparison. The logarithmic moment generating function then has, through weighted degree N, the form

$$\begin{aligned} \log \mathbb{E} \exp\left(\sum_{q \ge 0} t_q R_q \right) &= \frac{1}{2} \sum_{q \ge 0} (N - 2^q) \psi^{(q)}(2\sigma) t_q \\ &\quad + \frac{1}{8} \sum_{p,q \ge 0} \psi^{(p+q+1)}(2\sigma) t_p t_q - \sum_{r \ge 1} \frac{\psi^{(r-1)}(N\sigma)}{2^r r!} t_0^r. \end{aligned} \tag{2.11}$$

Here equality means coefficient equality only through that weighted degree, not an unrestricted identity of generating functions. The linear, quadratic and higher coefficients give (2.5)–(2.7). □

2.2 A finite formula for every retained ordinary moment

Let $\mathcal{P}_r$ be the set of partitions of $\{1, \dots, r\}$. For a block B define $c(B)$ by (2.5) when $|B| = 1$, by (2.6) when $|B| = 2$, and by (2.7) when $|B| \ge 3$, using the orders attached to the labels in B. Since every block satisfies (2.4) whenever the full list does, the ordinary moment is explicitly

$$\boxed{\mathbb{E} \prod_{i=1}^{r} R_{q_i} = \sum_{\pi \in \mathcal{P}_r} \prod_{B \in \pi} c(B), \qquad \sum_i (q_i + 1) \le N.} \tag{2.12}$$

This finite partition formula evaluates all the indicated integrals using only polygamma values at 2σ and $N\sigma$. It contains no unevaluated spectral derivative or auxiliary integral.

For positive orders define the centered statistics

$$Y_q = R_q - \frac{1}{2}(N - 2^q)\psi^{(q)}(2\sigma), \qquad q \ge 1.$$

All blocks of size other than two disappear from their centered moments. Consequently

Corollary 2.3 (Weighted Wick rule)**.** *For $q_i \ge 1$ and $\sum_i (q_i + 1) \le N$,*

$$\mathbb{E} \prod_{i=1}^{r} Y_{q_i} = \begin{cases} 0, & r \text{ odd}, \\ 4^{-r/2} \displaystyle\sum_{\text{pairings } \pi} \prod_{\{i,j\} \in \pi} \psi^{(q_i + q_j + 1)}(2\sigma), & r \text{ even}. \end{cases} \tag{2.13}$$

For example, for $n \ge 5$ the genuinely mixed fourth moment is

$$\mathbb{E}(Y_1^2 Y_3^2) = \frac{\psi'''(2\sigma)\psi^{(7)}(2\sigma)}{16} + \frac{\psi^{(5)}(2\sigma)^2}{8}. \tag{2.14}$$

At $\sigma = 1/4$ this is $33\pi^{12}/2$, using $\psi'''(1/2) = \pi^4$, $\psi^{(5)}(1/2) = 8\pi^6$, and $\psi^{(7)}(1/2) = 136\pi^8$. For $n \ge 2$ another mixed evaluation is

$$\mathbb{E}\big[(R_0 - \mathbb{E}R_0)(R_1 - \mathbb{E}R_1)(R_2 - \mathbb{E}R_2)\big] = 0, \tag{2.15}$$

whose total weight is $1 + 2 + 3 = 6$. The odd-order restriction, $S_{2a-1} = R_{2a-1}$, reads

$$\mathbb{E}S_{2a-1} = (n + 1 - 4^{a-1})\psi^{(2a-1)}(2\sigma), \qquad \operatorname{Cov}(S_{2a-1}, S_{2b-1}) = \tfrac{1}{4}\psi^{(2a+2b-1)}(2\sigma)$$

whenever, respectively, $a \le n + 1$ and $a + b \le n + 1$. Every higher cumulant vanishes when the sum of the corresponding a-indices is at most $n + 1$. This is the requested separation of the joint odd-polygamma coordinates.

2.3 New scalar integral evaluations

Set $n = 1$, $\sigma = 1/4$ and

$$\omega(x) = x\sinh(\pi x)|\Gamma(1/4 + ix/2)|^8, \quad z_x = 1/4 + ix/2,$$

$$g(x) = \Re\psi(z_x) + \gamma + 3\log 2, \qquad h(x) = \Re\psi'(z_x) - \pi^2/2.$$

Here $\int_0^\infty \omega(x)\,dx = 4\pi^5$. The theorem, whose weighted budget is $N = 4$, gives the exact ordinary integrals, using the standard special values in [7, §5.15],

$$\int_0^\infty \omega(x)\Re\psi''(z_x)\,dx = 0, \tag{2.16}$$

$$\int_0^\infty \omega(x)g(x)\Re\psi''(z_x)\,dx = \pi^9, \tag{2.17}$$

$$\int_0^\infty \omega(x)g(x)^2h(x)\,dx = 0, \tag{2.18}$$

$$\int_0^\infty \omega(x)g(x)^3\,dx = \pi^5\zeta(3), \tag{2.19}$$

$$\int_0^\infty \omega(x)g(x)^4\,dx = \frac{\pi^9}{15}. \tag{2.20}$$

For the last two, the centered digamma cumulants are

$$\kappa_2(R_0) = \frac{\pi^2}{12}, \quad \kappa_3(R_0) = \frac{\zeta(3)}{4}, \quad \kappa_4(R_0) = -\frac{\pi^4}{240}.$$

Thus the fourth central moment is $3(\pi^2/12)^2 - \pi^4/240 = \pi^4/60$; multiplication by $4\pi^5$ gives (2.20). The nonzero digamma cumulants explain why the Gaussian description belongs to the positive-order sector.

2.4 The finite range is essential

At degree $N + 1$, Newton's relation $e_{N+1} = 0$ in N variables creates a nonzero kernel:

$$\sum_{\lambda\vdash N+1} \frac{(-1)^{N+1-\ell(\lambda)}}{z_\lambda} p_{\lambda_1}\cdots p_{\lambda_{\ell(\lambda)}} = 0, \qquad z_\lambda = \prod_{j\ge1} j^{m_j} m_j!. \tag{2.21}$$

One therefore cannot compare all formal power-sum coefficients beyond the theorem's range. It is an algebraic failure of this extraction argument, not a proof that every first excluded cumulant is nonzero. The following limiting example supplies an actual obstruction to extending the source's single-trigamma Gaussian law universally.

Proposition 2.4 (A nonzero first excluded Gaussian moment)**.** *For $n = 1$ define the source-normalized statistic*

$$X_{1,\sigma} = \frac{2R_1 - 2\psi'(2\sigma)}{\sqrt{\psi'''(2\sigma)}}.$$

Then

$$\lim_{\sigma\downarrow0} \mathbb{E}X_{1,\sigma}^3 = \frac{2\sqrt{6}}{9} > 0. \tag{2.22}$$

In particular the Gaussian matching degree $n + 1$ cannot be increased to $n + 2$ uniformly over all $n \ge 1$ and all $\sigma > 0$. This proposition does not evaluate the general $(n + 2)$nd defect.

Proof. At $n = 1$, (2.2) is $Z_{1,\sigma} = 4\pi^2\Gamma(2\sigma)^6/\Gamma(4\sigma)$, and $\sigma^5 Z_{1,\sigma} \to \pi^2/4$. Set $x = 2\sigma y$. Using $z\Gamma(z) \to 1$ as $z \to 0$, the normalized density in the y variable tends pointwise to

$$q_0(y) = \frac{32}{\pi}\frac{y^2}{(1+y^2)^4}, \qquad y > 0.$$

Its integral is one by the beta integral. Since the approximating densities also integrate to one, Scheffé's lemma gives convergence in L^1. Also

$$\sigma^2 \Re\psi'(\sigma(1+iy)) \longrightarrow \Re(1+iy)^{-2}, \qquad \sigma^2|\psi'(\sigma(1+iy))| \le \sigma^2\psi'(\sigma),$$

and the upper bound tends to one. The density convergence and this uniform bound give convergence of every fixed moment of the scaled statistic. If $t = y^2/(1+y^2)$ then t has the Beta$(3/2, 5/2)$ law and

$$\Re(1+iy)^{-2} = f(t) = 1 - 3t + 2t^2.$$

The beta moment formula $\mathbb{E}t^j = (3/2)_j/(4)_j$ gives

$$\mathbb{E}f = \tfrac{1}{4}, \qquad \mathbb{E}(f - 1/4)^2 = \tfrac{3}{32}, \qquad \mathbb{E}(f - 1/4)^3 = \tfrac{1}{64}.$$

Finally $\sigma^2\psi'(2\sigma) \to 1/4$ and $\sigma^4\psi'''(2\sigma) \to 3/8$. Thus the normalized variable converges in moments to $\sqrt{32/3}(f - 1/4)$ and its third moment is $(32/3)^{3/2}/64 = 2\sqrt{6}/9$. □

2.5 A joint Gaussian consequence

Although the main conclusions are exact integral evaluations, the finite moment law also has a useful joint limit. Fix distinct positive orders $q_1, \dots, q_d$ and define

$$C(\sigma)_{ij} = \tfrac{1}{4}\psi^{(q_i+q_j+1)}(2\sigma).$$

This matrix is positive definite: the polygamma integral representation gives

$$v^T C(\sigma) v = \frac{1}{4}\int_0^\infty \left(\sum_i (-1)^{q_i} v_i t^{q_i}\right)^2 \frac{te^{-2\sigma t}}{1-e^{-t}}\,dt > 0$$

for every nonzero real v. Distinct monomial degrees ensure strictness. For any positive sequence σ_n, let

$$\mathbf{Z}_n = C(\sigma_n)^{-1/2}(R_{q_i} - \mathbb{E}R_{q_i})_{i=1}^d.$$

Every mixed moment of total ordinary degree k is exactly the corresponding standard d-variate Gaussian moment once $k(1 + \max_i q_i) \le 2n + 2$. This follows by expanding the deterministic linear change of coordinates and applying the weighted Wick rule. In particular $\mathbf{Z}_n$ converges in distribution to $N(0, I_d)$. For a proof, every fixed linear projection has eventually the exact Gaussian moments of each prescribed order; the second moments give tightness and higher even moments give uniform integrability. Moment determinacy of the normal law and the Cramér–Wold theorem finish the argument. No estimate uniform in σ_n is needed.

2.6 Independent verification

The accompanying program `verify_gamma_cumulants.py` performs 122 exact assertions. These verify the pair-sum identity at twelve orders and eight numbers of variables, verify the ranks of the power-sum substitution through the first excluded degree for up to five variables, and independently evaluate the beta moments and displayed quarter-parameter arithmetic. In particular the rank drops by one at weighted degree $N + 1$, as predicted by (2.21).

The numerical side uses continuous Cauchy–Binet in the opposite direction: it integrates one-dimensional moment entries with inserted products of polygammas, constructs a determinant of their finite Taylor jets, and extracts moments and cumulants by finite polynomial arithmetic. It does not use the Gamma quotient for the expectation side. Seventeen checks include the zero mean of R_2, the third and fourth centered digamma cumulants, mixed vanishing cumulants, and $\mathrm{Var}(R_2) = \mathrm{Cov}(R_1, R_3)$ at $n = 2$. The cases use quarter and non-quarter shifts. At 36 decimal digits, with 56 Gauss–Legendre nodes on each stated integration interval, the largest scaled residual is approximately 8.2×10^{-34}. A repeat at 46 decimal digits and 72 nodes per interval reduces the largest scaled residual to approximately 2.1×10^{-43}. A separate direct adaptive integral gives

$$\mathbb{E}X^3_{1,0.02} = 0.544394033224965748253671830809 2\ldots,$$
$$\mathbb{E}X^3_{1,0.005} = 0.544332155859557031675645658570 6\ldots.$$

consistent with the proved limit $2\sqrt{6}/9 = 0.544331053951817355154952016601 3\ldots$. These floating point residuals are independent representation checks, not certified error intervals or replacements for the analytic proofs.

3 Effective reconstruction at every fixed gap

The incoming ordered-boundary report [2] constructs a Gamma integral basis for the unequal-even-tail moment problem and proves eventual nonsingularity for each fixed gap. The subsequent gap-six report [3] makes one of those eventual statements effective and completes that ray by finite-field arithmetic. We now give a fully explicit threshold for *every* fixed gap. We then sharpen the estimates at gap eight and certify all remaining indices. In particular, the complete gap-eight family, which the inspected gap-six report explicitly leaves open, is resolved below.

The conclusions are statements about rational matrices on specified formal labels. They do not imply algebraic or rational independence of the evaluated zeta values. The contribution relative to the inspected sources is the explicit general threshold and the completed gap-eight ray; no historical-priority claim is made.

3.1 The inherited rational obstruction

Fix integers $m \geq 1$ and $b > m$, and put

$$p = 2b - 2m, \qquad q = 2b, \qquad x_n = n + \tfrac{1}{2}, \qquad A_v(n) = \zeta(v, n+1).$$

The meromorphic continuation of

$$D_{p,q}(s) = \sum_{n=0}^{\infty} A_p(n) A_q(n) x_n^{-s}$$

is regular at every nonpositive even argument. Indeed, the centered Hurwitz expansion has only powers x_n^{1-v-2h}, so the possible poles of the continued product sum are at the odd integers $3 - p - q - 2h$. Write $D_r = D_{p,q}(-2r)$.

The block coordinates from [2, 3] satisfy

$$D_r = \sum_{j=0}^{r} \frac{\binom{2r}{2j}}{2j+1} B_{2r-2j}(1/2) C_j, \tag{3.1}$$

$$C_j = \sum_{r=0}^{j} \binom{2j+1}{2r} 2^{-2(j-r)} D_r. \tag{3.2}$$

For $j \ge b$ they have the late coordinates

$$C_j = \sum_{k=0}^{b-1} a_{jk}\zeta(2k), \qquad \zeta(0) = -\tfrac{1}{2}.$$

Put $d = 2j+1$ and $e = d-p-q+2k+1$. The coefficient is zero if $e < 0$; otherwise it is

$$a_{jk} = B_e \left\{ \frac{\binom{d-q+1}{p-2k}}{d-q+1} + \frac{\binom{d-p+1}{q-2k}}{d-p+1} \right\}. \tag{3.3}$$

Each summand is omitted unless its lower binomial index is between 2 and the upper index. In particular the lower index zero is omitted. Define

$$M_{b,m} = (a_{b+r,k})_{0 \le r,k < b}.$$

The first $2b$ moment rows are expressed on the formal labels

$$C_0, \dots, C_{b-1}, \quad \zeta(0), \dots, \zeta(2b-2).$$

Their exact coefficient determinant is $\det M_{b,m}/(4b-1)!!$. Thus nonsingularity of $M_{b,m}$ is exactly the optimal-prefix question.

We recall the fixed-gap reduction in a notation which keeps the Bernoulli coefficients distinct from recurrence factors. Set

$$\begin{aligned} &\beta_n = \frac{2|B_{2n}|}{(2n)!}, \qquad n \ge 1, \\ &a_u = 0 \quad (1 \le u \le m), \qquad a_u = (2u)^{\underline{2m}} \quad (u > m), \\ &b_s = (2s-1)^{\underline{2m}}, \qquad d_s = 2(a_s + b_s) \quad (s > m). \end{aligned}$$

Here the underline means falling factorial. Define the rational fundamental array by

$$\begin{aligned} R_{u,j} &= \delta_{uj} && (1 \le u, j \le m), \\ R_{s,j} &= \frac{1}{d_s} \sum_{u=1}^{s-1} \beta_{s-u}(b_s + a_u) R_{u,j} && (s > m). \end{aligned} \tag{3.4}$$

Finally let

$$F_{b,m} = (R_{b+i,j})_{1 \le i,j \le m}, \qquad f_{b,m} = \det F_{b,m}.$$

The inherited complementary-minor identity is

$$\det M_{b,m} = \frac{(-1)^{b(b-1)/2}}{4^b b!} \left(\prod_{s=m+1}^{b+m} d_s \right) f_{b,m}. \tag{3.5}$$

Every factor in the product is strictly positive. Its normalization includes the omitted-index convention in (3.3). This identity follows by coefficient comparison in the null equation

$$(R^{(2m)} - R^{(2m)}(0)) \cot(z/2) + (R \cot(z/2))^{(2m)} = 0$$

and the complementary-minor identity for its lower-triangular coefficient operator; these inherited steps are reproduced in the ordered-boundary and gap-six sources.

3.2 Gamma realization and its signs

Put

$$W_m(x) = \left|\Gamma\left(\tfrac{1}{4} + \tfrac{ix}{2}\right)\right|^{4m},$$
$$P_j(x) = \sinh(\pi(m-j)x) \quad (1 \le j < m), \qquad P_m(x) = \tanh(\pi x),$$
$$\lambda_j = j\pi, \qquad A_m = 2^m(2\pi)^{2m}, \qquad a_j^* = \begin{cases} A_m/2, & j < m, \\ A_m, & j = m. \end{cases}$$

The incoming Gamma-basis theorem [2] gives

$$t_{n,j} = \frac{4}{(2n)!}\,\mathrm{Im}\int_0^\infty (x + i/2)^{2n} P_j(x) W_m(x)\,dx \tag{3.6}$$

and

$$(t_{b+i,j})_{i,j=1}^m = F_{b,m} T_m, \qquad T_m = (t_{n,j})_{n,j=1}^m, \qquad \operatorname{sgn}\det T_m = (-1)^{m(m-1)/2}. \tag{3.7}$$

All integrals converge absolutely. The null equation follows from the Gamma recurrence after two Fourier contours are shifted by $\pm i$; the sinh modes cross no poles, and the tanh mode supplies the two residues at $\pm i/2$ which give the subtracted derivative. The sign in (3.7) follows from continuous Cauchy–Binet: its integrand is a positive weight times the product of a Vandermonde in x_r^2 and a reversed Vandermonde in $\cosh(\pi x_r)$. These exact inputs, rather than numerical normalizations of the Gamma integrals, will be used below.

3.3 An explicit real-density estimate for arbitrary gaps

Lemma 3.1. *For $X = 2m$, every $x \ge X$, and $1 \le j \le m$,*

$$\frac{P_j(x) W_m(x)}{a_j^*} = x^{-m} e^{-\lambda_j x}(1 + \delta_{m,j}(x)), \qquad |\delta_{m,j}(x)| \le \frac{2m}{x^2}. \tag{3.8}$$

On $0 \le x \le X$ one has the elementary bound

$$0 \le P_j(x) W_m(x)/a_j^* < 2^{16m^2+12m}. \tag{3.9}$$

Proof. The complex Stirling remainder [7, §5.11(ii)], after retaining $1/(12z)$, gives

$$\log\Gamma(z) = (z - \tfrac{1}{2})\log z - z + \tfrac{1}{2}\log(2\pi) + \frac{1}{12z} + \rho(z), \qquad |\rho(z)| \le \frac{1}{90|z|^3}$$

for $z = 1/4 + ix/2$. Indeed the first omitted term has magnitude $1/(360|z|^3)$ and the secant factor is at most four in this closed right-half-plane sector. Let $t = 1/(2x)$ and

$$L_m(x) = \log \frac{W_m(x)}{A_m x^{-m} e^{-m\pi x}}.$$

Taking $4m$ times the real part gives exactly

$$L_m(x) = -\frac{m}{2}\log(1 + t^2) + m\left(\frac{\arctan t}{t} - 1\right) + \frac{4m}{3}\frac{t^2}{1 + t^2} + 4m\,\mathrm{Re}\,\rho(z).$$

Using $\log(1 + t^2) \le t^2$ and $0 \le 1 - \arctan(t)/t \le t^2/3$, for $x \ge 2$ we obtain

$$|L_m(x)| \le \frac{13m}{24x^2} + \frac{16m}{45x^3} \le \frac{259m}{360x^2} < \frac{3m}{4x^2}. \tag{3.10}$$

For $x \geq 2m$ this is at most $3/(16m) < 1$. The elementary inequality $|e^u - 1| \leq 2|u|$ for real $|u| \leq 1$ now gives an error at most $3m/(2x^2)$ for the normalized Gamma density. The remaining normalized sinh or tanh multiplier lies in $[0, 1]$ and differs from one by at most $2e^{-2\pi x} \leq 1/(9x^2)$. Combining these estimates proves (3.8).

Euler's integral gives $|\Gamma(1/4 + ix/2)| \leq \Gamma(1/4) < 5$. For $0 \leq x \leq 2m$, each multiplier is at most $e^{2\pi m(m-1)}$. Since $a_j^* > 1$, $\pi < 4$, $e < 4$, and $5 < 8$, the stated generous power-of-two bound follows. □

3.4 An effective shifted Laplace lemma

For $N > m + 1$ and $\lambda > 0$ define

$$S_N^{(r)}(\lambda) = \frac{\Gamma(N - m - r + 1)}{\lambda^{N-m-r+1}}, \qquad r = 0, 1, 2.$$

Let

$$C_m = \left\lceil 4m + \frac{3m(m+1)}{8} + 1 \right\rceil, \qquad N_0(m) = 4096(m+1)^2. \tag{3.11}$$

Lemma 3.2. *For every integer $N \geq N_0(m)$ and $1 \leq j \leq m$,*

$$\left| \frac{1}{a_j^*} \int_0^\infty (x + i/2)^N P_j(x) W_m(x)\, dx - e^{i\lambda_j/2} \left(S_N^{(0)}(\lambda_j) + \frac{im}{2} S_N^{(1)}(\lambda_j) \right) \right| < C_m S_N^{(2)}(\lambda_j). \tag{3.12}$$

Proof. Write $X = 2m$. On $[X, \infty)$ replace the normalized density by $x^{-m}e^{-\lambda x}$. Its absolute error is at most

$$2m \int_X^\infty |x + i/2|^N x^{-m-2} e^{-\lambda x}\, dx.$$

On $x \geq \sqrt{N}$ one has $|x + i/2|^N \leq x^N \exp(N/(8x^2)) < 2x^N$, so this part is at most $4mS_N^{(2)}$. On $X \leq x \leq \sqrt{N}$, the factor $2mx^{-m-2}$ is at most one; hence the remaining part is at most $\sqrt{N}(N+1)^{N/2}$.

For the model integral set $y = x + i/2$ and move the ray to $[X, \infty)$. Its integrand, after extracting $e^{i\lambda/2}$, is $y^{N-m}(1 - i/(2y))^{-m}e^{-\lambda y}$. No pole is crossed. On real $y \geq 2m$, Taylor's theorem applied to $(1-u)^{-m}$ gives

$$\left| (1 - i/(2y))^{-m} - 1 - \frac{im}{2y} \right| \leq \frac{3m(m+1)}{8y^2}.$$

For detail, $u = 1/(2y) \leq 1/(4m)$, and the positive-coefficient Taylor remainder is bounded by $m(m+1)u^2(1-u)^{-m-2}/2$. Also

$$(1-u)^{-m-2} \leq \exp\left(\frac{(m+2)u}{1-u} \right) \leq \exp\left(\frac{m+2}{4m-1} \right) < 3.$$

Thus the model remainder contributes at most $3m(m+1)S_N^{(2)}/8$.

The short vertical segment at X, the two retained integrals between 0 and X, and the original compact density integral are together bounded by $2^{20(m+1)^2}(3m)^N$. Indeed the compact density bound contributes at most $X2^{16m^2+12m}(X+1)^N$; the other three terms are bounded by $(2 + m/2)(X+1)^N$. The displayed larger constant covers their sum for every $m \geq 1$.

It remains to dominate

$$R_N = 2^{20(m+1)^2}(3m)^N + \sqrt{N}(N+1)^{N/2}$$

by $S_N^{(2)}$. Put $k = N - m - 2$. Since $\lambda_j < 4m$ and $k! \geq (k/3)^k$,

$$S_N^{(2)}(\lambda_j) > \frac{1}{4m}\left(\frac{k}{12m}\right)^k.$$

For $N \geq N_0(m)$, one has $k \geq 3N/4$ and $\sqrt{N+1} \leq (5/4)\sqrt{N}$. Consequently

$$\frac{\sqrt{N}(N+1)^{N/2}}{S_N^{(2)}} \leq N^{m+4}\left(\frac{20m}{\sqrt{N}}\right)^N < N^{m+4}2^{-N},$$
$$\frac{2^{20(m+1)^2}(3m)^N}{S_N^{(2)}} \leq 2^{20(m+1)^2}N^{m+3}\left(\frac{48m^2}{N}\right)^N < 2^{20(m+1)^2}N^{m+3}4^{-N}.$$

The elementary inequality $N^{m+4} \leq 2^{N-2}$ holds throughout this range. At $N_0(m)$ it follows from $\log_2 N_0(m) \leq 12 + 2m$ and $(m+4)(12+2m) < N_0(m) - 2$; it persists by induction because $N > 2(m+4)$ implies $(1 + 1/N)^{m+4} < 2$. The two ratios are therefore less than $1/4$ each, the second using also $N > 20(m+1)^2$. Thus $R_N < S_N^{(2)}/2$. Adding all errors proves (3.12). □

3.5 A closed, computable threshold for every fixed gap

Let $V_m = (j^{-2i})_{1 \leq i,j \leq m}$ and define the positive rational number

$$\delta_m = \frac{|\det V_m|}{\operatorname{perm} V_m}. \tag{3.13}$$

Here $\operatorname{perm} V_m$ is the sum of its $m!$ unsigned permutation products. Thus $0 < \delta_m \leq 1$, and its numerator is explicitly the absolute reciprocal-square Vandermonde product

$$|\det V_m| = \left(\prod_{j=1}^{m} j^{-2}\right)\prod_{j<k}(j^{-2} - k^{-2}).$$

Set

$$E_m = 8C_m + 2(m-1), \qquad \boxed{B(m) = \max\left\{2048(m+1)^2, 1 + \left\lceil \frac{2mE_m}{\delta_m} \right\rceil\right\}}. \tag{3.14}$$

Theorem 3.3 (Effective reconstruction on every fixed-gap ray)**.** *For every integer $m \geq 1$ and every $b \geq B(m)$,*

$$f_{b,m} \neq 0, \qquad \operatorname{sgn} f_{b,m} = (-1)^{\lfloor m^2/4 \rfloor}. \tag{3.15}$$

Consequently the first $q = 2b$ moments for $(p, q) = (2b - 2m, 2b)$ form an optimal formal prefix, and

$$\operatorname{sgn}\det M_{b,m} = (-1)^{b(b-1)/2 + \lfloor m^2/4 \rfloor}.$$

For any prescribed m, all possible exceptions lie in the explicitly specified finite interval $m + 1 \leq b < B(m)$.

Proof. Let $N_i = 2b + 2i$ for $1 \leq i \leq m$. At odd j, the leading phase in (3.12) is $\pm i$, and at even j it is ± 1. Thus the odd-mode leading term is

$$\sigma_j \frac{4a_j^*}{N_i!} S_{N_i}^{(0)}(\lambda_j), \qquad \sigma_j = \sin(j\pi/2),$$

with relative error bounded by

$$\frac{C_m \lambda_j^2}{(N_i - m)(N_i - m - 1)} < \frac{16C_m m^2}{(N_i - m - 1)^2}.$$

At even j the first nonzero imaginary term is

$$\sigma_j \frac{2ma_j^*}{N_i!} S_{N_i}^{(1)}(\lambda_j), \qquad \sigma_j = \cos(j\pi/2),$$

with relative error less than $8C_m/(N_i - m - 1)$. Retaining this second phase term is essential.

Extract the positive row factors

$$\frac{4\Gamma(N_i - m + 1)}{\Gamma(N_i + 1)} \pi^{-2i}$$

and the positive column factors $a_j^* \lambda_j^{-2b+m-1}$, multiplying each even column factor additionally by $m\lambda_j/(2(2b + 2 - m))$. Also extract its sign σ_j. The remaining array $E = (E_{ij})$ has the form

$$E_{ij} = j^{-2i}(1 + \eta_{ij}).$$

For an even column the extra row factor is

$$\rho_i = \frac{2b + 2 - m}{2b + 2i - m}, \qquad 1 - \rho_i \le \frac{2(m - 1)}{2b + 2 - m}.$$

Since $b \ge B(m) > m$ and $b \ge 2m^2$, the preceding estimates give

$$|\eta_{ij}| < \frac{E_m}{b} < \frac{\delta_m}{2m}. \tag{3.16}$$

For the odd columns use $16C_m m^2/b^2 \le 8C_m/b$; for the even columns use $|\rho_i(1 + \epsilon) - 1| \le 1 - \rho_i + |\epsilon|$.

Expanding the determinant into its signed permutation products,

$$|\det E - \det V_m| < \operatorname{perm} V_m \left\{ \left(1 + \frac{\delta_m}{2m}\right)^m - 1 \right\} < \operatorname{perm} V_m \, \delta_m = |\det V_m|.$$

The last strict inequality holds for $0 < \delta_m \le 1$: the binomial sum is bounded above by the corresponding infinite geometric sum $\delta_m/(2 - \delta_m) \le \delta_m$, with a strictly positive omitted tail. Hence $\det E$ has the sign of the reversed Vandermonde, namely $(-1)^{m(m-1)/2}$. The extracted column signs have product $\prod_j \sigma_j = (-1)^{\lfloor m^2/4 \rfloor}$. Dividing by the initial determinant in (3.7) cancels the reversed-Vandermonde sign and proves (3.15). Apply (3.5) to obtain the assertion about the original moment matrix. □

The formula is deliberately conservative and does not claim a least threshold. Its first values are

m	1	2	3	4	5	6
$B(m)$	8192	18432	32768	51200	635656	16146233.

The earlier complete gap-two, gap-four, and gap-six theorems are stronger on their individual rays. The advance here is an effective formula valid for arbitrary prescribed m. It does not assert that every one of the finite remaining matrices is nonsingular; the unrestricted all-pairs conjecture remains open.

4 The complete gap-eight theorem

For $m = 4$ sharper constants reduce the finite-completion threshold from 51200 to 8192. The next two estimates supply that reduction.

Lemma 4.1 (Gap-eight density and phase estimates)**.** *For $m = 4$, $x \ge 2$, and $1 \le j \le 4$,*

$$\frac{P_j(x)W_4(x)}{a_j^*} = x^{-4}e^{-\lambda_j x}(1+\delta_j(x)), \qquad |\delta_j(x)| < \frac{5}{x^2}.$$

For every integer $N \ge 16384$,

$$\left|\frac{1}{a_j^*}\int_0^\infty (x+i/2)^N P_j(x)W_4(x)\,dx - e^{i\lambda_j/2}\left(\frac{\Gamma(N-3)}{\lambda_j^{N-3}} + 2i\frac{\Gamma(N-4)}{\lambda_j^{N-4}}\right)\right| < 16\frac{\Gamma(N-5)}{\lambda_j^{N-5}}. \quad (4.1)$$

Proof. Equation (3.10) gives $|L_4(x)| < 3/x^2 \le 3/4$. The elementary Taylor-series bound $e^{3/4} < 17/8$ implies $|e^u - 1| \le (3/2)|u|$ for real $|u| \le 3/4$. Adding the normalized multiplier error $1/(9x^2)$ gives

$$|\delta_j(x)| < \left(\frac{9}{2} + \frac{1}{9}\right)x^{-2} < 5x^{-2}.$$

On $0 \le x \le 2$ the normalized density is less than 2^{80}, since $5^{16} < 2^{38}$, $e^{6\pi} < 3^{24} < 2^{39}$, and $a_j^* > 1$.

Repeat the proof of Lemma 3.2 with $X = 2$. The density error above $\sqrt{N}$ contributes less than $10\Gamma(N-5)/\lambda_j^{N-5}$. The model Taylor remainder has the sharper bound

$$\left|(1 - i/(2y))^{-4} - 1 - \frac{2i}{y}\right| < \frac{5}{y^2} \qquad (y \ge 2).$$

Indeed the positive series remainder divided by u^2, where $u = 1/(2y) \le 1/4$, is increasing and equals $1504/81$ at $u = 1/4$; its contribution is therefore $376/(81y^2) < 5/y^2$.

All compact and lower-range contributions are at most

$$R_N = 2^{82}3^N + \sqrt{N}(N+1)^{N/2}.$$

Now $\lambda_j \le 4\pi < 13$ and, with $k = N - 6$,

$$\frac{\Gamma(N-5)}{\lambda_j^{N-5}} > \frac{1}{13}\left(\frac{k}{39}\right)^k.$$

For every $N \ge 16384$,

$$\frac{39\sqrt{N+1}}{N-6} < \frac{1}{2}, \qquad \frac{117}{N-6} < \frac{1}{128}, \qquad N^7 \le 2^{N-2}.$$

The first inequality follows by squaring, checking the resulting quadratic at 16384, and noting its positive derivative thereafter; the second is immediate; the third holds at 16384 and persists by the same induction as in Lemma 3.2. Therefore the two normalized parts of R_N are bounded, respectively, by $2^{82}N^6 2^{-7N} < 1/4$ and $N^7 2^{-N} \le 1/4$. The total error is consequently less than $(10 + 5 + 1/2)$ times the displayed Gamma factor, proving (4.1). □

Proposition 4.2 (Certified positive tail at gap eight)**.** *For every integer $b \ge 8192$, $f_{b,4} > 0$.*

Proof. In (4.1), an odd column has relative error less than $2704/(N-5)^2$, and an even column has relative error less than $104/(N-5)$; these use only $\lambda_j < 13$. Perform the row and column normalizations in Theorem 3.3. The even-column row factors are now $\rho_i = (2b-2)/(2b+2i-4)$, with $1 - \rho_i \le 6/(2b-2)$. Hence the normalized entries satisfy

$$E_{ij} = j^{-2i}(1+\eta_{ij}), \qquad |\eta_{ij}| < \frac{1}{128} \qquad (b \ge 8192).$$

For the even columns use $6/(2b-2)+104/(2b-3)<110/(2b-3)<1/128$; the odd-column inequality is weaker still.

The four-by-four limiting matrix has exact values

$$\det V_4=\frac{175}{127401984},\qquad \operatorname{perm} V_4=\frac{157183}{4586471424},\qquad \frac{\det V_4}{\operatorname{perm} V_4}=\frac{6300}{157183}.$$

Moreover

$$\det V_4-\operatorname{perm} V_4\left\{(129/128)^4-1\right\}=\frac{357064449025}{1231171548132409344}>0.$$

Thus $\det E>0$. The extracted signs are $(+,-,-,+)$, whose product is positive, and $\det T_4>0$. Equation (3.7) proves $f_{b,4}>0$. □

4.1 An exact finite-field completion

The cotangent coefficients admit the elementary recurrence

$$\beta_1=\frac{1}{6},\qquad \beta_n=\frac{1}{2(2n+1)}\sum_{k=1}^{n-1}\beta_k\beta_{n-k}\quad (n\ge 2). \tag{4.2}$$

It follows by substituting the Taylor expansion of $(z/2)\cot(z/2)$ into its Riccati equation. Equations (4.2) and (3.4) can be computed exactly modulo the prime $\ell=2147483647=2^{31}-1$.

For $m=4$ and $5\le b\le 8191$ the largest fundamental row required is 8195, so only $\beta_1,\dots,\beta_{8194}$ are needed. Every denominator is invertible modulo ℓ. The denominators $2(2n+1)$ have their prime factors below ℓ, and

$$d_s=2\frac{(2s)!}{(2s-8)!}\frac{2s-4}{s}\qquad (s\ge 5) \tag{4.3}$$

has the same property throughout the stated finite range. Thus induction identifies every computed residue with the reduction of the exact rational entry.

Proposition 4.3 (Finite gap-eight certificate)**.** *For every integer $5\le b\le 8191$, the rational determinant $f_{b,4}$ has a nonzero reduction modulo* 2147483647.

Exact computational proof. The accompanying `verify_gap_eight.cpp` uses integer arithmetic only. It verifies primality by trial division through $\lfloor\sqrt{\ell}\rfloor$, generates (4.2), generates the four fundamental columns from (3.4), and evaluates each four-by-four determinant by finite-field Gaussian elimination. All 8187 determinant residues are nonzero; the full indexed list is supplied in `gap_eight_certificate.json`. As an independent exact check of the cotangent normalization, the program verifies all 8194 coefficients by multiplying $1-\frac{1}{2}\sum_{n\ge1}\beta_n t^n$ by the directly generated series of $\sin(\sqrt{t}/2)/(\sqrt{t}/2)$ and comparing with the direct series of $\cos(\sqrt{t}/2)$. The separate SymPy verifier builds rational Bernoulli numbers from their standard definition and checks the original matrix (3.3), the full bridge (3.5), and the finite-field residues independently for $5\le b\le 16$. These latter checks guard against a sign, factorial, or omitted-index error in passing from the source matrix to the recurrence. □

Theorem 4.4 (Every gap-eight pair has the optimal prefix)**.** *For every integer $b\ge 5$, the matrix $M_{b,4}$ is nonsingular. For $p=2b-8$ and $q=2b$, consequently,*

$$\begin{aligned}\operatorname{span}_{\mathbb{Q}}\{D_0,D_1,\dots\}&=\operatorname{span}_{\mathbb{Q}}\{D_0,\dots,D_{q-1}\}\\&=\operatorname{span}_{\mathbb{Q}}\{C_0,\dots,C_{b-1},\zeta(0),\dots,\zeta(2b-2)\}.\end{aligned}$$

The first q moment rows have formal rank q and coefficient determinant $\det M_{b,4}/(4b-1)!!$. *This is optimal for those q formal coordinates.*

Proof. Proposition 4.2 covers $b \geq 8192$, and Proposition 4.3 covers $5 \leq b \leq 8191$. Equation (3.5) proves nonsingularity of the source matrix. The triangular transforms (3.1)–(3.2) give the stated generation and formal-rank conclusions. □

The certificate proves nonvanishing at the finite indices; it does not determine their real signs. Direct rational checks give $f_{b,4} > 0$ for $5 \leq b \leq 16$, and the analytic theorem gives positivity for $b \geq 8192$. We make no positivity assertion for the intervening finite interval on the basis of the residue test.

4.2 An all-order family of absolutely convergent identities

The completed ray produces ordinary convergent evaluations of polygamma-tail products at every even pair separated by eight. Let $(\lambda_b, \dots, \lambda_{2b-1})$ be row zero of $M_{b,4}^{-1}$ and define

$$K_b(y) = \sum_{j=b}^{2b-1} \lambda_j y^{2j+1},$$

$$P_b(x) = K_b(x + \tfrac{1}{2}) - K_b(x - \tfrac{1}{2}) = \sum_{r=0}^{q-1} w_r x^{2r}.$$

For $0 \leq h \leq 4$ put

$$c_{v,0} = \frac{1}{v-1}, \qquad c_{v,h} = \frac{B_{2h}(1/2)}{(2h)!}(v)_{2h-1} \quad (h \geq 1), \qquad u_h = \sum_{e=0}^{h} c_{p,e} c_{q,h-e},$$

and set

$$S_b(x) = \sum_{h=0}^{4} \left(\sum_{e=0}^{h} u_e w_{q-1-h+e} \right) x^{8-2h}.$$

Every coefficient here is rational and specified by finite algebra.

Corollary 4.5 (Rational polygamma-product evaluations)**.** *For every $b \geq 5$, $p = 2b-8$, $q = 2b$, and $x_n = n + 1/2$,*

$$\boxed{\sum_{n=0}^{\infty} \left\{ \frac{P_b(x_n)}{(p-1)!(q-1)!} \psi^{(p-1)}(n+1)\psi^{(q-1)}(n+1) - S_b(x_n) \right\} = -\frac{1}{2}.} \tag{4.4}$$

The sum converges absolutely. The polynomial P_b has exact degree $2q-2$ and leading coefficient

$$w_{q-1} = \frac{4b(4b-1)f_{b-1,4}}{d_{b+4} f_{b,4}} \neq 0, \tag{4.5}$$

where at the initial index one uses

$$f_{4,4} = \frac{1}{1310557768119193412504125440000000}.$$

Proof. By construction, $\sum_{j=b}^{2b-1} \lambda_j C_j = \zeta(0) = -1/2$. The centered difference defining P_b has coefficients

$$w_r = \sum_{j=\max\{b,r\}}^{2b-1} \lambda_j \binom{2j+1}{2r} 2^{-2(j-r)}.$$

Thus (3.2) gives $\sum_r w_r D_r = -1/2$. The first five terms of the centered tail product are

$$A_p(n)A_q(n) = \sum_{h=0}^{4} u_h x_n^{2-p-q-2h} + O(x_n^{-p-q-8}).$$

Multiplication by P_b leaves precisely S_b as its nonnegative powers, with the remainder $O(x_n^{-2})$. The analytically continued sums of $1, x_n^2, \dots, x_n^8$ vanish because $\zeta(-2h, 1/2) = 0$. Therefore the ordinary absolutely convergent sum in (4.4) equals that moment combination. Finally, for even v, $A_v(n) = \psi^{(v-1)}(n+1)/(v-1)!$.

For $b \geq 6$, deleting the last row and first column of $M_{b,4}$ and reindexing j, k, p, q downward by $1, 1, 2, 2$, respectively, gives exactly $M_{b-1,4}$. Cramer's rule and (3.5) give $\lambda_{2b-1} = 4b f_{b-1,4}/(d_{b+4} f_{b,4})$. The centered difference multiplies the top coefficient by $4b-1$, proving (4.5). At $b = 5$ the same relation is verified directly with the displayed rational auxiliary determinant. Theorem 4.4 proves all other required nonvanishing. □

4.3 A concrete pair beyond the previous small-order coverage

For $(p, q) = (14, 22)$, namely $b = 11$, the construction gives an even polynomial P_{11} of degree 42 and a subtraction polynomial S_{11} of degree 8. The leading coefficient is

$$[x^{42}]P_{11}(x) = \frac{45225337352233233630380}{10060146014319869101877031}.$$

All 22 coefficients of P_{11} and all five coefficients of S_{11} are supplied as exact rationals in `gap_eight_14_22_example.json`. The short generator `create_gap_eight_example.py` checks the inverse row against the original Bernoulli matrix and verifies the entire subtraction polynomial symbolically. Thus the example specifies a directly usable ordinary sum, without printing several pages of large rational coefficients. The finite definitions above are the full exact formula for every other pair as well.

5 Stieltjes collisions at different powers of the small parameter

The unequal-rate collision theorem in the latest gap-six report [3, Section 4 and research question 10] assumes $a = \lambda\epsilon$ and $b = \mu\epsilon$ with fixed positive rates. Its proof explicitly leaves different powers of ϵ open. Here we resolve that question for every pair of positive monomial scales and every fixed pair of Stieltjes indices and argument derivatives. The additional constants occur at explicitly classified rational ratios of the exponents. In particular, the product of the two singular parts can now contribute a constant.

The local subtraction argument is a classical asymptotic method; related generalized Stieltjes-transform methods appear in [8]. The contribution here is the complete resonance classification and its evaluated contact formula for this family, rather than a claim to invent finite-part asymptotics.

5.1 A compact identity before the general theorem

Write γ for Euler's constant, and put

$$D(a, b) = \frac{\log(b/a)}{b-a} = \int_0^\infty \frac{dx}{(x+a)(x+b)}, \qquad a, b > 0. \tag{5.1}$$

The value at $a = b$ is $1/a$. A derivative in the first argument is written D_a; explicitly,

$$D_a(a, b) = -\frac{1}{a(b-a)} + \frac{\log(b/a)}{(b-a)^2}. \tag{5.2}$$

Theorem 5.1 (A two-scale ordered digamma identity)**.** *Fix $\lambda, \mu, \alpha, \beta > 0$, and set $a = \lambda\epsilon^{\alpha}$, $b = \mu\epsilon^{\beta}$. Then, as $\epsilon \downarrow 0$,*

$$\boxed{\int_0^1 \psi'(x+a)\psi(x+b)\,dx = D_a(a,b) + \frac{\psi(1+b)}{a} + \zeta(2)\log\frac{b}{a} + o(1).} \tag{5.3}$$

Every term on the left is an ordinary convergent integral. The subtraction on the right is exact to a remainder tending to zero along every indicated monomial path.

Proof. Let $h(x) = \psi(1+x)$, so that $\psi(x) = -1/x + h(x)$. Expand the product and integrate the term $h(x+b)/(x+a)^2$ by parts. The difference between the two sides of (5.3), without $o(1)$, is

$$\begin{aligned} &- D_a(1+a, 1+b) - \frac{h(1+b)}{1+a} + \zeta(2)\log\frac{1+a}{1+b} \\ &\quad + \int_0^1 \left\{ \frac{h'(x+b) - \zeta(2)}{x+a} - \frac{h'(x+a) - \zeta(2)}{x+b} \right\} dx + \int_0^1 h'(x+a)h(x+b)\,dx. \end{aligned} \tag{5.4}$$

In the middle integral replace $h'(x+b)$ and $h'(x+a)$ by $h'(x)$. The two errors are $O(b(1 + |\log a|) + a(1 + |\log b|))$. Since $h'(x) - \zeta(2) = O(x)$, the remaining difference tends to zero by dominated convergence. The first, second, and last terms in (5.4) tend respectively to

$$\frac{1}{2}, \qquad \gamma - 1, \qquad \frac{1}{2}(\psi(2)^2 - \psi(1)^2) = \frac{1}{2} - \gamma.$$

They cancel. A positive power of ϵ times any fixed power of $|\log\epsilon|$ tends to zero, proving the statement. □

For a function with a power–logarithm expansion, let CT_ϵ denote the coefficient of $\epsilon^0(\log\epsilon)^0$. It does not mean the ordinary limit of a divergent integral. Let $\mathbf{1}_{\mathcal{E}}$ denote the indicator of the stated condition.

Corollary 5.2 (An explicit resonance law)**.** *Suppose $q = \alpha/\beta > 1$, and set $j = 2/(q-1)$. For $I(\epsilon)$ equal to the integral in (5.3),*

$$\begin{aligned} \mathrm{CT}_\epsilon\, I = \zeta(2)\log\frac{\mu}{\lambda} &+ \mathbf{1}_{q\in\mathbb{Z}_{\geq 2}}(-1)^{q+1}\frac{\mu^q}{\lambda}\zeta(q+1) \\ &+ \mathbf{1}_{j\in\mathbb{Z}_{\geq 1}}\frac{\lambda^j}{\mu^{j+2}}\left((j+1)\log\frac{\mu}{\lambda} - 1\right). \end{aligned} \tag{5.5}$$

A term whose indicator is zero is omitted, including its integer-indexed notation. In particular, at $\lambda = \mu = 1$,

$$\mathrm{CT}_\epsilon\, I = \begin{cases} -1 - \zeta(3), & q = 2, \\ -1 + \zeta(4), & q = 3, \\ (-1)^{q+1}\zeta(q+1), & q \in \mathbb{Z},\ q \geq 4, \\ -1, & q = 1 + 2/j,\ j \geq 3, \\ 0, & q \notin \mathbb{Z}_{\geq 2} \text{ and } 2/(q-1) \notin \mathbb{Z}_{\geq 1}. \end{cases} \tag{5.6}$$

Proof. Use the convergent expansion $\psi(1+b) = -\gamma + \sum_{\ell\geq 1}(-1)^{\ell+1}\zeta(\ell+1)b^\ell$ for $|b| < 1$. In $\psi(1+b)/a$, exponent zero requires $\ell\beta = \alpha$. If $\rho = a/b$, the other singular term has the convergent expansion

$$D_a(a,b) = -\frac{1}{ab} + \frac{1}{b^2}\sum_{k\geq 0}\rho^k\left((k+1)\log\frac{b}{a} - 1\right). \tag{5.7}$$

Here exponent zero requires $k(\alpha-\beta) = 2\beta$. Extracting the constant in $\log\epsilon$ proves the result. □

These are new exact values of specified asymptotic coefficients. They also give ordinary renormalized limits by subtracting the finitely many divergent powers and logarithms in (5.7) and the displayed digamma series. For example, if $L = \log \epsilon$, the first two cases are

$$\lim_{\epsilon\downarrow 0}\Bigg[\int_0^1 \psi'(x+\epsilon^2)\psi(x+\epsilon)dx + \epsilon^{-3} + (1+\gamma+L)\epsilon^{-2} \\ +(1+2L-\zeta(2))\epsilon^{-1} + (3+\zeta(2))L\Bigg] = -1-\zeta(3), \tag{5.8}$$

$$\lim_{\epsilon\downarrow 0}\Bigg[\int_0^1 \psi'(x+\epsilon^3)\psi(x+\epsilon)dx + \epsilon^{-4} + \gamma\epsilon^{-3} + (1+2L-\zeta(2))\epsilon^{-2} \\ +\zeta(3)\epsilon^{-1} + (4+2\zeta(2))L\Bigg] = -1+\zeta(4). \tag{5.9}$$

The convergence is established analytically by the theorem; the numerical diagnostics in the package test the signs and the independent expansions.

5.2 Definitions for arbitrary Stieltjes indices

We fix the standard convention

$$Z(u,x) = \zeta(1+u,x) - \frac{1}{u} = \sum_{m\ge 0}\frac{(-1)^m}{m!}\gamma_m(x)u^m, \qquad \gamma_0(x) = -\psi(x). \tag{5.10}$$

Argument derivatives are indicated by a parenthesized superscript. Put

$$\begin{aligned} P_r(u) &= (-1)^r(1+u)_r, \qquad A_r(u) = \frac{(1+u)_r}{r!}, \\ Z_r(u,x) &= \partial_x^r Z(u,x), \qquad g_h(u) = \partial_x^h Z(u,x)\Big|_{x=1}. \end{aligned} \tag{5.11}$$

These are holomorphic near $u=0$. The Hurwitz recurrence [7, §25.11] gives the exact splitting

$$Z_r(u,x) = P_r(u)x^{-r-1-u} + Z_r(u,1+x). \tag{5.12}$$

In particular $g_0(u) = \zeta(1+u) - 1/u$ and, for $h \ge 1$,

$$g_h(u) = (-1)^h(1+u)_h\zeta(1+u+h). \tag{5.13}$$

Thus every derivative of g_h is a finite combination of ordinary zeta derivatives and generalized harmonic numbers; for $h=0$ its coefficients are the ordinary Stieltjes constants.

The reference finite part is

$$F_{m,n}^{r,t} = \mathrm{CT}_{\delta\downarrow 0}\int_\delta^1 \gamma_m^{(r)}(x)\gamma_n^{(t)}(x)\,dx. \tag{5.14}$$

It uses the coordinate δ exactly as displayed. An explicit holomorphic germ $\mathcal{F}_{r,t}(u,v)$ for these coefficients can be defined entirely by an ordinary convergent integral. With $R = r+t+1$, set

$$\begin{aligned} \mathcal{R}(u,v;x) &= Z_r(u,x)Z_t(v,x) - P_r(u)P_t(v)x^{-R-1-u-v} \\ &\quad - P_r(u)\sum_{k=0}^{r}\frac{g_{t+k}(v)}{k!}x^{k-r-1-u} - P_t(v)\sum_{k=0}^{t}\frac{g_{r+k}(u)}{k!}x^{k-t-1-v}, \end{aligned} \tag{5.15}$$

$$\begin{aligned} \mathcal{F}_{r,t}(u,v) &= \int_0^1 \mathcal{R}(u,v;x)dx - \frac{P_r(u)P_t(v)}{R+u+v} \\ &\quad - P_r(u)\sum_{k=0}^{r-1}\frac{g_{t+k}(v)}{k!(r-k+u)} - P_t(v)\sum_{k=0}^{t-1}\frac{g_{r+k}(u)}{k!(t-k+v)}. \end{aligned} \tag{5.16}$$

Empty sums vanish. Then $F^{r,t}_{m,n} = (-1)^{m+n}\partial_u^m\partial_v^n\mathcal{F}_{r,t}(0,0)$. The critical powers $x^{-1}\log^h x$ contribute zero to the constant on $(0,1)$; this explains their omission from the two rational sums in (5.16). This construction agrees with the inherited convention [3, 5].

For example, those sources give

$$\begin{gathered} F^{0,0}_{0,0} = 2\gamma_1 - 2\zeta(2), \quad F^{1,0}_{0,0} = \zeta(2), \quad F^{1,1}_{0,0} = 2\zeta(3) + 4\zeta'(3), \\ F^{0,0}_{1,0} = \tfrac{3}{2}\gamma_2 + 2\gamma\zeta(2) - \zeta(3). \end{gathered} \tag{5.17}$$

These baseline evaluations are credited to the previous work. The new result evaluates every change caused by the different scales.

5.3 The complete constant formula

Assume henceforth that $\alpha > \beta > 0$ and define

$$a = \lambda\epsilon^\alpha, \quad b = \mu\epsilon^\beta, \quad d = \alpha - \beta, \quad \tau = \lambda/\mu, \quad R = r + t + 1. \tag{5.18}$$

The opposite ordering follows by exchanging the factors. Introduce the finite set

$$\mathscr{L}_r(\alpha,\beta) = \{(k,\ell) : 0 \le k < r,\ \ell \in \mathbb{Z}_{\ge 1},\ \beta\ell = \alpha(r-k)\}. \tag{5.19}$$

It has at most r members, including at rational ratios with large numerator or denominator. Define the following removable germ:

$$\begin{aligned} \mathcal{C}_{r,t}(u,v;\lambda,\mu;\alpha,\beta) &= (-1)^r g_{r+t}(v)\frac{\lambda^{-u} - A_r(u)}{u} \\ &\quad + (-1)^r \sum_{(k,\ell)\in\mathscr{L}_r(\alpha,\beta)} \frac{\mu^\ell\lambda^{k-r-u}}{\ell!}(1+u)_{r-k-1}\, g_{t+k+\ell}(v). \end{aligned} \tag{5.20}$$

The first quotient is removable since $A_r(0) = 1$.

There is one possible singular–singular contact. If $j = \beta R/d$ is a positive integer, set

$$\begin{aligned} \mathcal{H}_j(u,v) &= \frac{(-1)^t\mu^{-R-v}\tau^j}{u\Gamma(1+v)} \\ &\quad \times \left\{ \frac{\lambda^{-u}\Gamma(R+j+v)}{(1-u)_j} - \frac{\mu^{-u}\Gamma(1-u)\Gamma(R+j+u+v)}{j!} \right\}. \end{aligned} \tag{5.21}$$

The numerator vanishes identically at $u = 0$, so this is a holomorphic two-variable germ. Define $\mathcal{H}_* = \mathcal{H}_j$ when $\beta R/d = j \in \mathbb{Z}_{\ge 1}$, and $\mathcal{H}_* = 0$ otherwise. The quotient must be removed before taking derivatives.

Theorem 5.3 (All-index monomial collision constants)**.** *For every $m,n,r,t \in \mathbb{Z}_{\ge 0}$ and all positive parameters in (5.18), the integral*

$$I^{r,t}_{m,n}(\epsilon) = \int_0^1 \gamma_m^{(r)}(x+a)\gamma_n^{(t)}(x+b)\,dx$$

has a power–logarithm asymptotic expansion through its constant. Its exact constant is

$$\boxed{\begin{aligned} \mathrm{CT}_\epsilon\, I^{r,t}_{m,n} = F^{r,t}_{m,n} + (-1)^{m+n}\partial_u^m\partial_v^n\big[&\mathcal{C}_{r,t}(u,v;\lambda,\mu;\alpha,\beta) \\ &+\mathcal{C}_{t,r}(v,u;\mu,\lambda;\beta,\alpha) + \mathcal{H}_*(u,v)\big]_{u=v=0}. \end{aligned}} \tag{5.22}$$

Every correction in this formula is a finite expression in logarithms, ordinary zeta values and derivatives, Stieltjes constants at one, and generalized harmonic numbers. There is no new unevaluated spectral constant in the correction.

There are precisely three possible mechanisms for extra constants:

$$\beta\ell = \alpha(r-k), \qquad \alpha\ell = \beta(t-k), \qquad j(\alpha-\beta) = \beta(r+t+1). \tag{5.23}$$

The two first equations have the finite ranges specified above; the last has at most one admissible integer j. If α/β is irrational, all three resonance contributions vanish. Only the first, critical quotient in each $\mathcal{C}$ remains. These are selection rules for an asymptotic coefficient, not statements of discontinuity of the original positive-shift integral.

5.4 A two-shift expansion with analytic coefficients retained

We prove the theorem in two steps. Keeping the analytic coefficients at their actual small shifts is the useful replacement for a fixed-ratio Taylor truncation. Define

$$\mathcal{B}_{r,t}(u,v;a,b) = P_r(u)P_t(v)\int_0^\infty (x+a)^{-r-1-u}(x+b)^{-t-1-v}\,dx. \tag{5.24}$$

This is an ordinary absolutely convergent integral for u, v near zero. It also has the elementary incomplete-beta representation used below.

Lemma 5.4 (Uniform germ along monomial paths)**.** *For fixed exponents and rates as in* (5.18)*, locally uniformly on some complex neighborhood of* $(u,v) = (0,0)$*,*

$$\begin{aligned}\int_0^1 Z_r(u,x+a)Z_t(v,x+b)dx &= \mathcal{F}_{r,t}(u,v) + \mathcal{B}_{r,t}(u,v;a,b)\\ &+ (-1)^r\sum_{k=0}^{r-1}(1+u)_{r-k-1}a^{k-r-u}Z_{t+k}(v,1+b) + (-1)^r g_{r+t}(v)\frac{a^{-u}-A_r(u)}{u}\\ &+ (-1)^t\sum_{k=0}^{t-1}(1+v)_{t-k-1}b^{k-t-v}Z_{r+k}(u,1+a) + (-1)^t g_{r+t}(u)\frac{b^{-v}-A_t(v)}{v} + o(1).\end{aligned} \tag{5.25}$$

All fixed spectral derivatives of the remainder tend to zero.

Proof. Use (5.12). The singular–singular product on $(0,1)$ equals (5.24) minus its tail on $(1,\infty)$; the latter tends uniformly to $P_r(u)P_t(v)/(R+u+v)$. For the first mixed product, use Taylor's theorem in x at the point $1+b$:

$$Z_t(v,1+b+x) = \sum_{k=0}^{r}\frac{Z_{t+k}(v,1+b)}{k!}x^k + O(x^{r+1}).$$

The remainder has a bound uniform for small b and for v in a fixed disk. After multiplication by the singular factor it is bounded by $Cx^{-\eta}$, with $\eta<1$, independent of the relative sizes of a, b. Dominated convergence therefore applies. For $k<r$ the beta integral gives

$$\begin{aligned}P_r(u)\int_0^1 x^k(x+a)^{-r-1-u}dx &= (-1)^r k!(1+u)_{r-k-1}a^{k-r-u}\\ &\quad - \frac{P_r(u)}{r-k+u} + o(1).\end{aligned}$$

For $k=r$ its removable version is

$$P_r(u)\int_0^1 x^r(x+a)^{-r-1-u}dx = (-1)^r r!\frac{a^{-u}-A_r(u)}{u} + o(1).$$

One can first take $\Re u>0$, subtract the tail, and continue; the divided difference is uniformly holomorphic near zero. Replacing the coefficient of the critical term at $1+b$ by its value at

one costs $O(b\,a^{-\eta}(1+|\log a|))$ on a small disk. Choose $\eta>0$ so small that $\beta-\alpha\eta>0$; the error then tends to zero. For the exchanged term also impose $\alpha-\beta\eta>0$. The upper endpoints and the remaining integrals give exactly (5.16). This proves local uniformity. Cauchy's integral formula on a slightly larger disk justifies every prescribed spectral derivative. □

The Taylor expansion

$$Z_{t+k}(v,1+b)=\sum_{\ell\ge0} g_{t+k+\ell}(v)\frac{b^\ell}{\ell!}$$

now shows that a noncritical mixed term has base exponent $\alpha(k-r)+\beta\ell$. Taking it far enough to have a positive remainder exponent is always a finite task. Exponent zero gives $\mathscr{L}_r(\alpha,\beta)$; the critical quotient contributes its constant in $\log\epsilon$ by replacing a^{-u} with λ^{-u}. This proves exactly the two $\mathcal{C}$ terms.

5.5 Pairing the singular branches before differentiation

Put $A=r+1+u$, $B=t+1+v$, $C=A+B-1=R+u+v$ and $\rho=a/b$. For $0<a<b$, the substitution $y=(x+a)/(x+b)$ gives

$$\frac{\mathcal{B}_{r,t}}{P_r(u)P_t(v)}=(b-a)^{-C}\left\{\frac{\Gamma(1-A)\Gamma(C)}{\Gamma(B)}-\int_0^\rho y^{-A}(1-y)^{C-1}\,dy\right\}. \tag{5.26}$$

At generic nonintegral A the last integral is understood by its analytic continuation; the original integral in (5.24) fixes the branch and the removable values. Expansion of this incomplete beta function, or the usual hypergeometric connection formula [7, §15.8], gives two normally convergent series for $0<\rho<1$:

$$\begin{aligned}\frac{\mathcal{B}_{r,t}}{P_r(u)P_t(v)}&=b^{-C}\frac{\Gamma(1-A)}{\Gamma(B)}\sum_{j\ge0}\frac{\Gamma(C+j)}{j!}\rho^j\\&\quad+a^{1-A}b^{-B}\sum_{k\ge0}(-1)^k(B)_k\frac{\Gamma(A-1-k)}{\Gamma(A)}\rho^k.\end{aligned} \tag{5.27}$$

The individual series have poles at $u=0$. Their sum is holomorphic, and the poles must be paired at equal base exponents.

The $k<r$ terms of the second branch are holomorphic and have base exponents $-\alpha r-\beta(t+1)+k(\alpha-\beta)<0$. For each $j\ge0$, pair the first-branch term indexed by j with the second-branch term indexed by $k=r+j$. Their common base exponent is

$$E_j=-\beta R+j(\alpha-\beta). \tag{5.28}$$

After the Gamma reflection and recurrence identities, the paired term is $\epsilon^{E_j}\mathcal{H}_j(u,v;L)$, where $L=\log\epsilon$ and

$$\begin{aligned}\mathcal{H}_j(u,v;L)&=\frac{(-1)^t\mu^{-R-v}\tau^j\epsilon^{-\beta v}}{u\Gamma(1+v)}\\&\quad\times\left\{\frac{\lambda^{-u}\epsilon^{-\alpha u}\Gamma(R+j+v)}{(1-u)_j}-\frac{\mu^{-u}\epsilon^{-\beta u}\Gamma(1-u)\Gamma(R+j+u+v)}{j!}\right\}.\end{aligned} \tag{5.29}$$

For each fixed L, this is holomorphic in (u,v). Its constant in L after any fixed spectral differentiation is obtained by setting $L=0$ before differentiating; the result is (5.21).

Choose a finite integer J with $(J+1)(\alpha-\beta)>\beta R$, then choose the spectral radius $\eta>0$ small enough that also $(J+1)(\alpha-\beta)-\beta R-(\alpha+\beta)\eta>0$. On this sufficiently small spectral

disk the tail $j > J$ in the paired series tends uniformly to zero: the coefficient growth is at most polynomial in j, while ρ^j is geometric. The divided difference in u introduces only a fixed logarithmic factor, which can be absorbed by shrinking the spectral disk. Thus only $E_j = 0$ can contribute a constant, and this is exactly the condition defining $\mathcal{H}_*$. Together with the preceding mixed-term analysis this proves Theorem 5.3 and also provides all divergent terms through a finite truncation of (5.25) and (5.27).

5.6 Harmonic and zeta evaluations of the new contacts

At spectral degree zero, the singular contact is particularly simple. With $H_0 = 0$,

$$\boxed{\mathcal{H}_j(0,0) = (-1)^t \frac{(R+j-1)!}{j!} \frac{\lambda^j}{\mu^{R+j}} \left(\log \frac{\mu}{\lambda} + H_j - H_{R+j-1} \right).} \tag{5.30}$$

This follows by differentiating the numerator of (5.21) once in u. The derivative of $\log(1-u)_j^{-1}$ is H_j at zero; the derivative of $\log\Gamma(1-u) + \log\Gamma(R+j+u)$ is H_{R+j-1}. Formula (5.30) recovers the second resonance family in Corollary 5.2.

For arbitrary spectral degrees, the finite Taylor computations use

$$\begin{aligned} \log \frac{j!}{(1-u)_j} &= \sum_{h \geq 1} \frac{H_j^{(h)}}{h} u^h, \\ \log\Gamma(N+u) - \log\Gamma(N) &= \psi(N)u + \sum_{h \geq 2} \frac{(-1)^h}{h} (\zeta(h) - H_{N-1}^{(h)}) u^h. \end{aligned} \tag{5.31}$$

Truncate these convergent expansions at the required degree and exponentiate, equivalently using complete Bell polynomials. This proves the stated finite constant alphabet constructively.

For completeness, the singular contact can be evaluated without leaving even a Gamma derivative in the answer. Put $M = R + j$ and $D = m + n + 1$, and form two finite polynomials:

$$\begin{aligned} Q_D(u,v) = & -u \log\lambda - v \log\mu + \sum_{h=1}^{D} \frac{H_j^{(h)}}{h} u^h + \sum_{h=1}^{D} \frac{(-1)^{h+1} H_{M-1}^{(h)}}{h} v^h, \\ T_D(u,v) = & -(u+v)\log\mu + \sum_{h=1}^{D} \frac{(-1)^{h+1} H_{M-1}^{(h)}}{h} (u+v)^h \\ & + \sum_{h=2}^{D} \frac{\zeta(h)}{h} \left[u^h + (-1)^h \left((u+v)^h - v^h \right) \right]. \end{aligned} \tag{5.32}$$

Define the finite exponential polynomial $\mathcal{E}_D(X) = \sum_{k=0}^{D} X^k/k!$. Then the evaluated contact is

$$\begin{aligned} & (-1)^{m+n} \partial_u^m \partial_v^n \mathcal{H}_j(0,0) \\ & \quad = \frac{(-1)^{m+n+t} m! n! (M-1)!}{j!} \frac{\lambda^j}{\mu^M} [u^{m+1} v^n] \left\{ \mathcal{E}_D(Q_D) - \mathcal{E}_D(T_D) \right\}. \end{aligned} \tag{5.33}$$

The notation $[u^a v^b]$ extracts a coefficient from an ordinary finite polynomial. To prove the formula, divide the two Gamma products in (5.21) by their common constant $(M-1)!/j!$. Their logarithms, including the amplitude factors, have Taylor polynomials Q_D and T_D. Euler's constant cancels in each. Both polynomials have zero constant term, so terms beyond the Dth power of their exponential cannot affect the requested coefficient. Division by the removable u is exactly extraction at u^{m+1}. This gives a finite expression in harmonic numbers and ordinary zeta values at every specified pair of Stieltjes indices. The two mixed contacts are obtained just

as finitely by multiplying the polynomials $(1+u)_k$, the exponential of $-u\log\lambda$, and the zeta jets in (5.13).

There is an especially transparent first-Stieltjes example. Take $r=t=0$, so $R=1$. A singular contact occurs exactly when $\alpha/\beta = 1+1/j$, $j\ge 1$. Direct expansion of (5.21) yields

$$-\partial_u \mathcal{H}_j(0,0) = \frac{\lambda^j}{\mu^{j+1}}\Big[\zeta(2) - H_j^{(2)} + \frac{(H_j-\log\mu)^2 - (H_j-\log\lambda)^2}{2}\Big]. \tag{5.34}$$

Consequently, for $a=\epsilon^{j+1}$ and $b=\epsilon^j$,

$$\boxed{\mathrm{CT}_\epsilon \int_0^1 \gamma_1(x+\epsilon^{j+1})\gamma_0(x+\epsilon^j)dx = \frac{3}{2}\gamma_2 + 2\gamma\zeta(2) - \zeta(3) + \zeta(2) - H_j^{(2)}.} \tag{5.35}$$

The new term is the ordinary trigamma tail $\zeta(2)-H_j^{(2)} = \psi'(j+1)$. Thus all positive integer trigamma tails appear as exact Stieltjes collision contacts.

This family also has a classical polylogarithmic generator: for $|z|<1$,

$$\sum_{j\ge 1}(\zeta(2)-H_j^{(2)})z^j = \frac{\zeta(2)z - \mathrm{Li}_2(z)}{1-z}. \tag{5.36}$$

Its ordinary primitive can be normalized at zero using

$$\int_0^z \frac{1}{t}\sum_{j\ge 1}(\zeta(2)-H_j^{(2)})t^j dt = 2\,\mathrm{Li}_{2,1}(z) - \mathrm{Li}_3(z) + (\mathrm{Li}_2(z)-\zeta(2))\log(1-z). \tag{5.37}$$

Here $\mathrm{Li}_{2,1}(z) = \sum_{n>k\ge 1} z^n/(n^2k)$, on its analytic branch at zero. The generating identity is an elementary interchange of absolutely convergent sums. The primitive identity is checked by differentiation using $\frac{d}{dz}\mathrm{Li}_{2,1}(z) = \mathrm{Li}_{1,1}(z)/z = \log^2(1-z)/(2z)$ together with the corresponding product derivatives; an independent quadrature check is recorded with the computational artifacts. In fact classical polylogarithms suffice: for $0<z<1$, the same primitive equals

$$2\zeta(3) - \mathrm{Li}_3(z) - 2\,\mathrm{Li}_3(1-z) + \log(1-z)\,\mathrm{Li}_2(1-z). \tag{5.38}$$

To verify this expression directly, differentiate and use Euler's dilogarithm reflection identity; its limit at zero is zero. At $z=1$ it gives the endpoint identity $\sum_{j\ge1}(\zeta(2)-H_j^{(2)})/j = \zeta(3)$. The elementary generating and primitive identities are classical Euler-sum consequences; their role here is to evaluate the newly identified contact family, not to claim those consequences as independently new polylogarithm identities.

5.7 What the different-scale theorem changes

The fixed-rate source remains correct with its stated hypotheses. It must not be extrapolated by substituting an ϵ-dependent rate in a remainder proved only for fixed rates. Two changes are essential: high Taylor coefficients at one can survive when b^ℓ/a^{r-k} has exponent zero, and the two singular Frobenius branches can jointly have exponent zero. Both effects are present in (5.6); the first-index law (5.35) isolates the second effect without argument derivatives.

The theorem does not assert uniformity as the exponent ratio approaches a resonance while ϵ also varies. In that transition, a factor ϵ^{E_j} must be retained rather than replaced by its constant coefficient. Formula (5.29) provides the exact local term for such a further analysis.

6 A Barnes-integral companion

This section records evaluated consequences of the classical first Barnes lemma. The underlying integral is not new: it is NIST DLMF Equation 5.13.3 [7]. The present contribution is the selection and complete differentiation of the centered fourth moment and the pure trigamma correlation. No historical priority claim is made for these specializations. These ordinary integrals are distinct from the higher-dimensional Gamma density used in the recent *Gamma Normalizations and Colored Moments* report.

For complex parameters of positive real part, put

$$W(a,b,c,d;x) = \Gamma(a+ix)\Gamma(b+ix)\Gamma(c-ix)\Gamma(d-ix), \tag{6.1}$$

$$F(a,b,c,d) = \frac{\Gamma(a+c)\Gamma(a+d)\Gamma(b+c)\Gamma(b+d)}{\Gamma(a+b+c+d)}. \tag{6.2}$$

Barnes' first lemma states

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} W(a,b,c,d;x)\,dx = F(a,b,c,d). \tag{6.3}$$

Every finite parameter derivative can be passed under this integral. Indeed, on a compact parameter set inside the four right half-planes, Stirling's formula gives exponential decay $e^{-2\pi|x|}$ times a fixed polynomial in $1+|x|$; differentiating adds only a fixed power of $1+\log(2+|x|)$. This also proves that all integrals below are ordinary absolutely convergent integrals.

6.1 Centered mixed moments

For $\sigma > 0$ define

$$p_\sigma(x) = \frac{\Gamma(4\sigma)}{2\pi\Gamma(2\sigma)^4}|\Gamma(\sigma+ix)|^4, \qquad \int_{\mathbb{R}} p_\sigma(x)\,dx = 1. \tag{6.4}$$

Write $\mathbb{E}_\sigma$ for integration against this density, and put

$$m_\sigma = 2\psi(2\sigma) - \psi(4\sigma), \qquad U_\sigma(x) = \psi(\sigma+ix) - m_\sigma. \tag{6.5}$$

The following formulas involve only ordinary polygamma values:

$$\mathbb{E}_\sigma U_\sigma = 0, \tag{6.6}$$

$$\mathbb{E}_\sigma U_\sigma^2 = -\psi'(4\sigma), \tag{6.7}$$

$$\mathbb{E}_\sigma |U_\sigma|^2 = \psi'(2\sigma) - \psi'(4\sigma), \tag{6.8}$$

$$\mathbb{E}_\sigma\big(U_\sigma|U_\sigma|^2\big) = -\psi''(4\sigma), \tag{6.9}$$

$$\mathbb{E}_\sigma |U_\sigma|^4 = \psi'(4\sigma)^2 + 2\big(\psi'(2\sigma) - \psi'(4\sigma)\big)^2 - \psi'''(4\sigma). \tag{6.10}$$

The left side of (6.9) is real by the change $x \mapsto -x$.

Proof. Set $L = \log F$. At $a=b=c=d=\sigma$, all first logarithmic derivatives equal m_σ. Subtract this linear term by replacing $F/F(\sigma,\sigma,\sigma,\sigma)$ with

$$\widetilde{F}(a,b,c,d) = e^{-m_\sigma(a+b+c+d-4\sigma)}\frac{F(a,b,c,d)}{F(\sigma,\sigma,\sigma,\sigma)}.$$

Its single-slot integral insertions are U_σ in the slots a,b and $\overline{U}_\sigma$ in c,d. The first derivatives of $\log\widetilde{F}$ vanish. For distinct slots the remaining derivatives needed here are

$$\begin{aligned} L_{ab} = L_{cd} = q &= -\psi'(4\sigma), \\ L_{ac} = L_{ad} = L_{bc} = L_{bd} = p &= \psi'(2\sigma) - \psi'(4\sigma), \\ L_{abc} = L_{abd} = L_{acd} = L_{bcd} &= -\psi''(4\sigma), \\ L_{abcd} &= -\psi'''(4\sigma). \end{aligned}$$

The derivatives of $\widetilde{F} = e^{\log \widetilde{F}}$ of orders two and three give the first four assertions. Its *abcd* derivative is $L_{ab}L_{cd} + L_{ac}L_{bd} + L_{ad}L_{bc} + L_{abcd}$, because every partition containing a singleton has zero contribution. This is $q^2 + 2p^2 - \psi'''(4\sigma)$ and proves (6.10). □

6.2 Eliminating all digamma insertions from a trigamma correlation

Theorem 6.1. *For $\sigma > 0$ one has*

$$\mathbb{E}_\sigma \psi'(\sigma + ix) = 2\psi'(2\sigma), \tag{6.11}$$

$$\mathbb{E}_\sigma \left|\psi'(\sigma + ix)\right|^2 = 4\psi'(2\sigma)^2 + \psi'''(2\sigma). \tag{6.12}$$

More generally, if $\Re u, \Re v > 0$ and $s = u + v$, then

$$\begin{aligned} &\int_{\mathbb{R}} \Gamma(u + ix)^2 \Gamma(v - ix)^2 \psi'(u + ix)\psi'(v - ix)\, dx \\ &\qquad = 2\pi \frac{\Gamma(s)^4}{\Gamma(2s)} \big(4\psi'(s)^2 + \psi'''(s)\big). \end{aligned} \tag{6.13}$$

Proof. Use the constant-coefficient operators

$$D_+ = \partial_a^2 - \partial_a\partial_b, \qquad D_- = \partial_c^2 - \partial_c\partial_d.$$

After setting $a = b = u$ the first operator inserts exactly $\psi'(u + ix)$, since the two copies of $\psi(u + ix)^2$ cancel. The second operator inserts $\psi'(v - ix)$ after $c = d = v$.

For an explicit right-side calculation, center as above, now at $a = b = u, c = d = v$; the common first logarithmic derivative is $2\psi(s) - \psi(2s)$. Put

$$\begin{gathered} A = \psi'(s), \quad B = \psi'(2s), \quad A_3 = \psi'''(s), \quad B_3 = \psi'''(2s), \\ d = 2A - B, \quad q = -B, \quad p = A - B. \end{gathered}$$

Centering does not affect $D_+D_-\widetilde{F}$ at coalescence: the extra terms contain $\partial_a - \partial_b$ or $\partial_c - \partial_d$, which vanish by the respective symmetries. The required derivatives are

$$\begin{aligned} \widetilde{F}_{aacc} &= d^2 + 2p^2 + A_3 - B_3, \\ \widetilde{F}_{aacd} = \widetilde{F}_{abcc} &= dq + 2p^2 - B_3, \\ \widetilde{F}_{abcd} &= q^2 + 2p^2 - B_3. \end{aligned}$$

Consequently

$$D_+D_-\widetilde{F} = (d - q)^2 + A_3 = 4A^2 + A_3.$$

Multiplying by $2\pi F(u, u, v, v)$ proves (6.13); setting $u = v = \sigma$ proves (6.12). The second-order calculation $\widetilde{F}_{aa} - \widetilde{F}_{ab} = d - q = 2A$ gives (6.11). □

The unequal-shift formula can equally be written using $\psi'(z) = \zeta(2, z)$ and $\psi'''(z) = 6\zeta(4, z)$. Thus it is a Gamma-weighted Hurwitz-zeta convolution with two independently chosen argument shifts. The only shift invariant on the right is their sum.

6.3 Explicit half- and quarter-argument evaluations

At $\sigma = 1/2$, the Gamma reflection formula makes $p_{1/2}(x) = (\pi/2)\operatorname{sech}^2(\pi x)$ and $U_{1/2}(x) = \psi(1/2 + ix) + \gamma + 1$. Equations (6.8), (6.10), and (6.12) become

$$\int_{\mathbb{R}} \frac{|\psi(1/2+ix)+\gamma+1|^2}{\cosh^2(\pi x)}\,dx = \frac{2}{\pi}, \tag{6.14}$$

$$\int_{\mathbb{R}} \frac{|\psi(1/2+ix)+\gamma+1|^4}{\cosh^2(\pi x)}\,dx = \frac{2}{\pi}\left(9 - \frac{\pi^2}{3} - \frac{7\pi^4}{180}\right), \tag{6.15}$$

$$\int_{\mathbb{R}} \frac{|\psi'(1/2+ix)|^2}{\cosh^2(\pi x)}\,dx = \frac{16\pi^3}{45}. \tag{6.16}$$

Here the cancellation of Euler's constant is exact after centering.

At $\sigma = 1/4$, put $V(x) = \psi(1/4 + ix) + \gamma + 4\log 2$. The normalization is $p_{1/4}(x) = |\Gamma(1/4 + ix)|^4/(2\pi^3)$, and one obtains the equally short evaluations

$$\int_{\mathbb{R}} |\Gamma(1/4+ix)|^4 |V(x)|^2\,dx = \frac{2\pi^5}{3}, \tag{6.17}$$

$$\int_{\mathbb{R}} |\Gamma(1/4+ix)|^4 |V(x)|^4\,dx = \frac{11\pi^7}{30}, \tag{6.18}$$

$$\int_{\mathbb{R}} |\Gamma(1/4+ix)|^4 |\psi'(1/4+ix)|^2\,dx = 4\pi^7. \tag{6.19}$$

The special-value reductions use DLMF 5.15.2 and 5.15.3.

7 Source audit, integration decisions, and further research

The source anchor for this continuation is ProveIt revision `03112a7d3b91` [1]. The audit concerns the incoming passages used in the proofs and the manuscript's relevant status records. It is not a certification of every result in the repository. The three main advances have the following precise relation to that source state.

7.1 What the present results change

1. **Joint Gamma–polygamma moments.** The Gamma-normalization report's `sections/02_gaussian.tex` proves the paired deformation and the single-trigamma Gaussian moments through degree $n + 1$; research question Q2 in `sections/06_audit_research.tex` asks for the joint polygamma moments [4]. Theorem 2.1 separates all joint cumulants through total weight $2n + 2$, with weight $q + 1$ assigned to R_q. It includes digamma and even polygamma orders. Formula (2.12) then evaluates every retained ordinary moment, and Corollary 2.3 gives the centered positive-order Wick rule. This answers the joint-separation question in the stated weighted range; the beyond-range classification remains open. The nonzero small-σ skewness limit in Proposition 2.4 supplies an obstruction relevant to the same report's Q1, but does not evaluate the general first excluded trigamma moment. The underlying Gustafson integral remains classical [6]; its Gamma normalizer and the paired law retain their incoming attribution.

2. **Effective reconstruction and the complete gap-eight ray.** Section 8 of the ordered-boundary report gives eventual nonsingularity at each fixed gap; Section 9, research question 1, asks for an effective threshold and finite completion [2]. Theorem 3.3 supplies an explicit computable $B(m)$ for every fixed half-gap m. The later gap-six report's `sections/05_`

`audit_and_questions.tex`, question Q1, explicitly asks for complete gap-eight coverage [3]. Theorem 4.4 proves this for every $(p, q) = (2b - 8, 2b)$, $b \geq 5$, using a proved analytic tail and an exact finite-field completion. Corollary 4.5 turns that result into ordinary convergent polygamma-product identities. The unrestricted all-pairs prefix conjecture remains open. The finite certificate proves nonvanishing, and does not prove positivity at every intervening finite index.

3. **Stieltjes collisions at different monomial scales.** The gap-six report's Section 4 treats fixed positive rates $a = \lambda\epsilon$, $b = \mu\epsilon$; its question Q10 asks for $a = \lambda\epsilon^{\alpha}$, $b = \mu\epsilon^{\beta}$ with $\alpha \neq \beta$ [3]. Theorem 5.3 evaluates the constant for every positive monomial pair and every fixed pair of Stieltjes indices and argument derivatives. Its proof also specifies all divergent terms through that constant by finite truncation. The three resonance conditions in (5.23) identify the additional contacts, including the singular–singular contact. The earlier lattice-contact report's Q9 asked only for fixed unequal rates [5]; that problem was already answered by the gap-six report and is not reclaimed here. General nonmonomial paths and three independently shifted factors remain outside the present theorem. The trigamma-tail contact family (5.35) and its polylogarithmic generator and primitive are explicit evaluated consequences of the new scale classification.

The Barnes-integral companion supplies additional ordinary integral evaluations obtained by explicitly differentiating a classical Barnes lemma [7]. It should be credited in that form. A bounded literature search and comparison with the inspected incoming material support the stated continuation choices; they do not establish global historical novelty.

7.2 Errors, status corrections, and essential qualifications

No newly false inherited mathematical theorem was found in the specific dependencies used for these results. The changes above are proved extensions and research-status updates. They should not be presented as corrections to formulas which the sources stated correctly. In particular, the incoming Gamma report already restricts its finite Gaussian law, and the gap-six report already distinguishes finite-field nonvanishing from a real-sign theorem.

One source-list reconciliation is necessary. The gap-six report's Q4 and the ordered-boundary report's question 2 ask for evaluations of K_m. The later Gamma-normalization report already supplies them [4, 3, 2]. These questions should be marked answered by that predecessor, rather than counted among the present results. The repeated README paragraph about the 1,032-input first-letter-X family, rank 1,008 and kernel dimension 24 remains present at the pinned revision. Earlier incoming reports already identified this editorial duplication. Removing its second occurrence would be an inherited editorial cleanup, not a new mathematical erratum.

Three conventions are material to the proofs. The Gamma moment classification retains its total-weight restriction: beyond that range, Newton relations prevent unrestricted comparison of power-sum coefficients. The Bernoulli reconstruction matrix retains the omission of a lower binomial index zero; evaluating that entry as $\binom{n}{0} = 1$ changes the problem. The Stieltjes constant is taken in the displayed small parameter, with the fixed-coordinate finite part kept distinct from the path-dependent constant. Spectral pole pairs must be combined before their removable values and derivatives are taken. These are integration requirements, not newly discovered mistakes in the cited sources.

The manuscript's short S_6 and revised S_8 candidates remain open; the old rejected S_8 candidate remains rejected. None of the Gamma moment, reconstruction, or collision theorems implies those polylogarithmic period identities. Numerical proximity and rank in a specified formal relation space do not supply their missing exact equalities. The verification artifacts here likewise distinguish exact finite certificates from floating-point diagnostics. The analytic

arguments justify convergence, continuation and coefficient extraction.

7.3 Further questions directed toward exact identities

1. **Evaluate the first excluded polygamma cumulants.** At total weight $2n+3$ and above, determine the relations induced by $e_{2n+3}, e_{2n+4}, \dots$ and identify additional integral identities that separate the remaining cumulants. The immediate scalar target is the exact $(n+2)$nd trigamma Gaussian defect at general n, σ. The proved $n = 1$ small-shift limit is a normalization constraint on any proposed formula.

2. **Develop joint moment formulas for unequal Gamma clusters.** Allow several distinct base shifts with specified multiplicities. Determine which mixed polygamma moments can be separated by independent deformations within each cluster, how their finite coordinate relations change when clusters merge, and which new ordinary integrals arise at the first unresolved degrees. A useful answer would give explicit polygamma or finite hypergeometric coefficients, rather than only a differentiated generating integral.

3. **Find closed reconstruction rows and identities in b.** The gap-eight theorem specifies P_b by a rational inverse matrix. Find finite-sum descriptions or recurrences for its coefficients and for the corresponding subtraction polynomial. Decide which inverse rows yield especially short zeta-valued identities. The exact leading coefficient and the ordinary sum in (4.4) provide tests for every proposed recurrence.

4. **Replace fixed-gap finite completion by structural algebra.** Find a recurrence invariant, factorization, or appropriate variation-diminishing statement that proves nonvanishing for an entire gap ray without checking every exceptional index separately. Such a theorem must respect the known early sign behavior. Connecting this structure with the fixed-smaller-order results is a concrete route toward the unrestricted unequal-even prefix problem.

5. **Compute collision constants for nonmonomial paths.** Take $a = \lambda\epsilon^{\alpha}A(\epsilon)$ and $b = \mu\epsilon^{\beta}B(\epsilon)$, with prescribed analytic or power–logarithm expansions and nonzero leading coefficients. Determine the finite jets of A, B that can affect a specified constant and derive its exact transformation law under a change of the small parameter. The monomial resonance rules identify the base exponents with which these additional coefficients interact.

6. **Evaluate three-factor Stieltjes collision contacts.** For three independently shifted derivatives, classify contributions from every singular subset and pair all intersecting spectral poles before differentiating. Seek a finite evaluated contact formula in logarithms, generalized harmonic numbers and zeta derivatives, and identify precisely where multivariable polylogarithms or other new functions become necessary.

7. **Generate and integrate the higher contact families.** For fixed Stieltjes and derivative orders, sum the integer-resonance contacts against z^j and determine their polylogarithmic alphabet. The trigamma-tail generator and its anchored primitive are the first case. Compute further Euler primitives, ordinary or finite-part endpoint values, and character-weighted versions; determine which reduce to zeta products and which retain genuine multiple polylogarithms.

8. **Construct an explicit functional route to S_6 or revised S_8.** If one of the new generators or integral deformations is to inform these period problems, first exhibit its exact map to the canonical residual, including all products and branch constants. A new functional or iterated-integral identity outside the tested relation spaces would constitute progress; an unrelated rank theorem or additional decimal agreement would not.

8 Reproducibility and the role of computation

The package contains the complete article as one standalone TeX file, the modular source, its compiled PDF, all verification programs, and the recorded results. The exact source revision and the inherited inputs are identified in the bibliography. No running service, unpublished data, or repository mutation is required to reproduce the calculations. The mathematical assertions are separated into three types of evidence.

Analytic proofs. The joint-cumulant theorem is proved by a classical Gamma integral and finite polynomial injectivity. The effective reconstruction tail is proved by an explicit two-term phase expansion and a rational determinant margin. The monomial collision theorem is proved by uniform local subtraction, finite selection of exponents, and holomorphic cancellation of paired singular branches. The ordinary Barnes evaluations follow by differentiated integral identities. The limit and sum convergence arguments are part of those proofs.

Exact finite computation. The C++ program `verify_gap_eight.cpp` uses only unsigned integer arithmetic and arithmetic in the prime field $\mathbb{F}_{2147483647}$. All 8187 determinant residues for $5 \le b \le 8191$ are explicitly recorded. The verifier checks the prime and every needed denominator and independently checks all 8194 cotangent coefficients against the sine and cosine series. An independent replay of the compiled program reproduced the complete certificate exactly. The SymPy bridge program separately checks the original Bernoulli matrices and their rational determinant reduction for $5 \le b \le 16$, and the concrete $(14, 22)$ sum has its own exact generator. The finite-field certificate proves rational nonvanishing; it does not estimate a real determinant or deduce its sign.

Independent symbolic and numerical diagnostics. The Gamma program performs 122 exact algebra assertions and 17 integral checks. The latter extract cumulants from determinants of independently integrated moment entries; they do not substitute the same Gamma quotient on both sides. The two precision runs and their residuals are described in Section 2. The Stieltjes compiler implements the finite contact rule using exact SymPy expressions. Its test program performs 72 exact assertions, including comparisons with derivatives of the elementary logarithmic kernel, the first-index dilogarithmic contact, and the inherited fixed-rate law. Five nonreal spectral checks compare the paired branch series with the original convergent kernel integral; three further checks compare the classical primitive with direct integration. At 64 digits the largest scaled spectral residual is below 8.2×10^{-56}.

The separate `identity_audit.py` evaluates ordered collision remainders and six Barnes moment formulas at 90 decimal digits. The Barnes residuals are below 3×10^{-91}. The collision diagnostics deliberately include a nonresonant irrational exponent ratio, for which a small positive exponent makes convergence slow. That example is recorded with its vanishing-term acceleration rather than mislabeled as a rapidly convergent check. All floating-point residuals are diagnostics, not rigorous numerical error intervals. The exact theorem does not depend on their size.

How to reproduce. The `README.md` gives the precise build and verification commands. The minimal numerical dependencies are Python 3, SymPy, and mpmath; the certificate additionally needs a C++17 compiler. The TeX uses standard mathematical packages and compiles with pdfLaTeX through `latexmk`. Internal analytical audit notes identify the delicate normalizations and remainder arguments. These are separate derivations within this research workflow and are not a claim of external peer review or proof-assistant certification.

References

[1] V. Reshetnikov, *ProveIt*, polylogarithm manuscript and incoming research reports, repository revision `03112a7d3b91efdd3c00b6e595cf0ffa4694d2dc`, 11 October 2026 (repository commit date), inspected 12 October 2026. Pinned manuscript; pinned incoming directory.

[2] ProveIt incoming report, *Ordered Boundary Identities*, 11 October 2026, archive `ProveIt_Ordered_Boundary_Identities_2026-10-11.zip` in [1]. The inherited fixed-gap recurrence, Gamma basis, complementary-minor bridge, and eventual nonsingularity theorem.

[3] ProveIt incoming report, *Gap-Six Harmonic Determinants*, 12 October 2026, archive `ProveIt_Gap_Six_Harmonic_Determinants_2026-10-12.zip` in [1]. See `sections/01_gap_six.tex`, `sections/04_collisions.tex`, and `sections/05_audit_and_questions.tex`.

[4] ProveIt incoming report, *Gamma Normalizations and Colored Moments*, 12 October 2026, archive `ProveIt_Gamma_Normalizations_and_Colored_Moments_2026-10-12.zip` in [1]. See `sections/02_gaussian.tex` and `sections/06_audit_research.tex`, especially research question 2.

[5] ProveIt incoming report, *Lattice Contacts and Mixed Zeta Identities*, 11 October 2026, archive `ProveIt_Lattice_Contacts_and_Mixed_Zeta_Identities_2026-10-11.zip` in [1].

[6] S. E. Derkachov and A. N. Manashov, *Spin Chains and Gustafson's Integrals*, Journal of Physics A: Mathematical and Theoretical **50** (2017), 294006. doi:10.1088/1751-8121/aa749a; arXiv:1611.09593v2. The symplectic Barnes integral used here is equation (2).

[7] F. W. J. Olver et al., eds., *NIST Digital Library of Mathematical Functions*, https://dlmf.nist.gov/, accessed 12 October 2026. In particular §5.11 (Gamma asymptotics and remainders), §5.13 (Barnes integrals), §5.15 (polygamma functions), §15.8 (hypergeometric transformations), and §25.11 (Hurwitz zeta functions).

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[9] M. A. Shpot and R. B. Paris, *Integrals of products of Hurwitz zeta functions via Feynman parametrization and two double sums of Riemann zeta functions*, arXiv:1609.05658v2 (2016; revised 2020).